

HCDE 511

Information Visualization

Winter 2026

Week

5



Week

Visual Perception Pt. 2

Break

Tasks & Interactions

Actions & Targets activity

Interactions

Break

Interactions Activity

Next Week Preview

Week 1

Intro
Data
encoding &
mapping

2

Graphical
excellence
& integrity
Critiquing

3

Fundamen-
tal graphs
Working
with data

4

Visual
perception

5

Visual
perception,
Pt 2.
Tasks &
inter-
actions

6

Collabora-
tive
critique

7

Multi-
variate
analysis

8

Trees &
networks

9

Animation,
context,
story-
telling

10

Final pre-
sentations

Finals

A1 Good/bad

A2 Static vis

A3 Exploratory analysis

Project proposal

Rev. prop.

Final project deliverable

Final project writeup

Projects

What's new?

What did you learn?

What changed?

Visual Perception

Part 2

gestalt principles

similarity

proximity

closure

continuation

figure and ground

symmetry and order

common fate

good figure or simplicity
(prägnanz)

gestalt principles

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closure

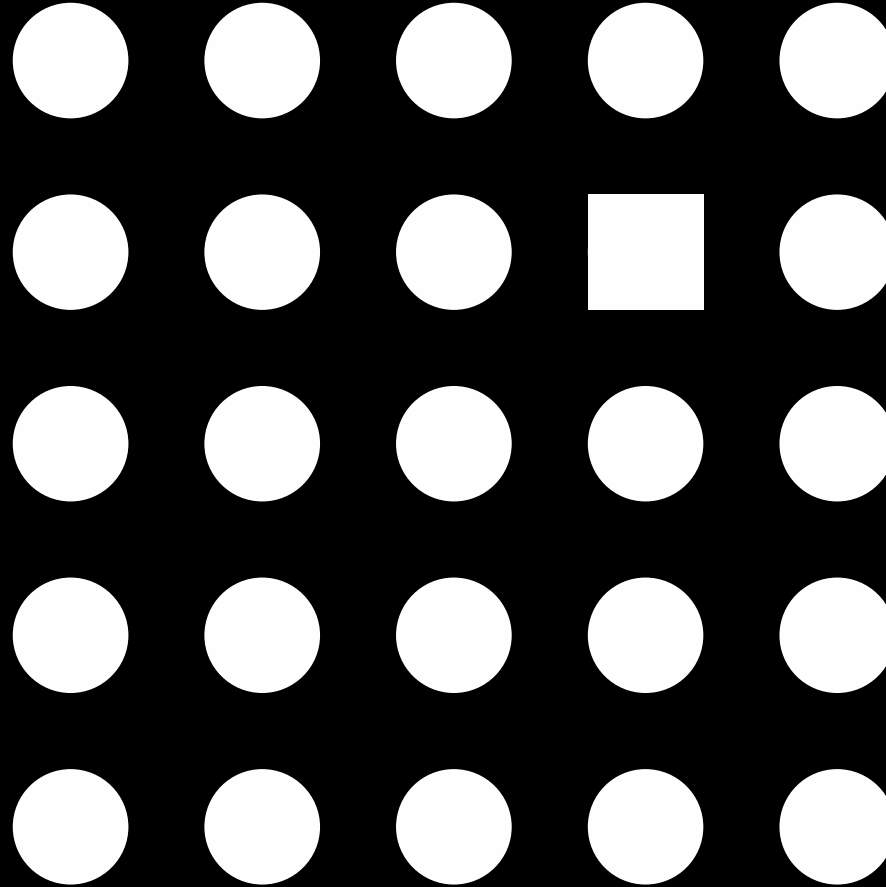
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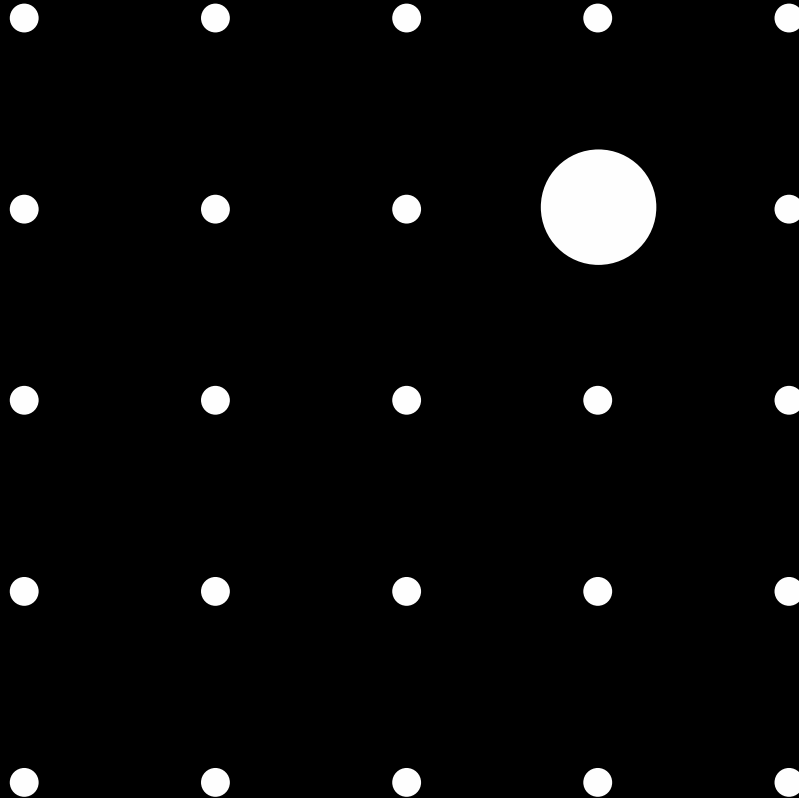
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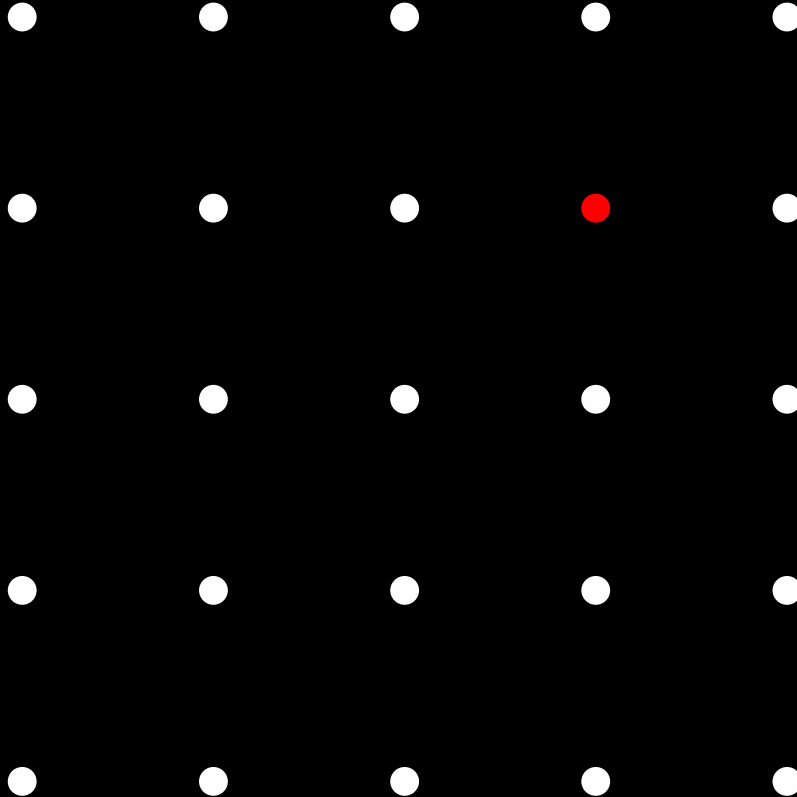
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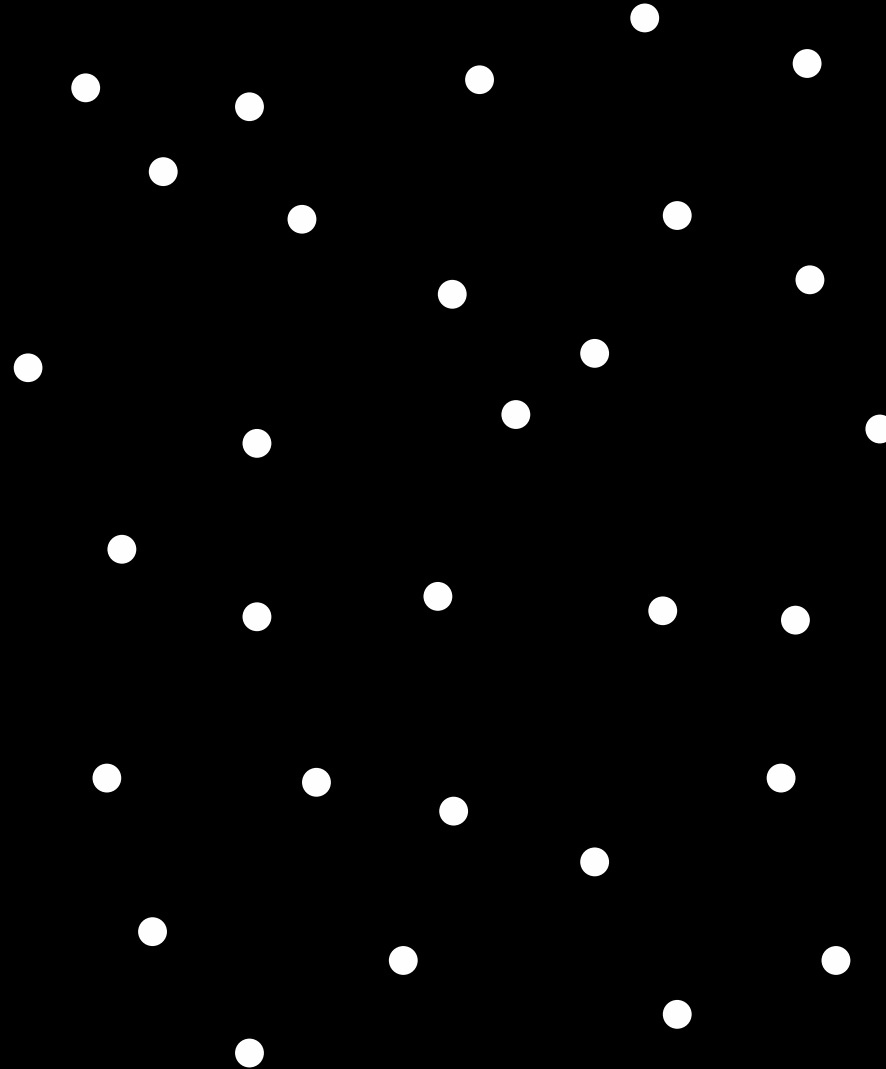
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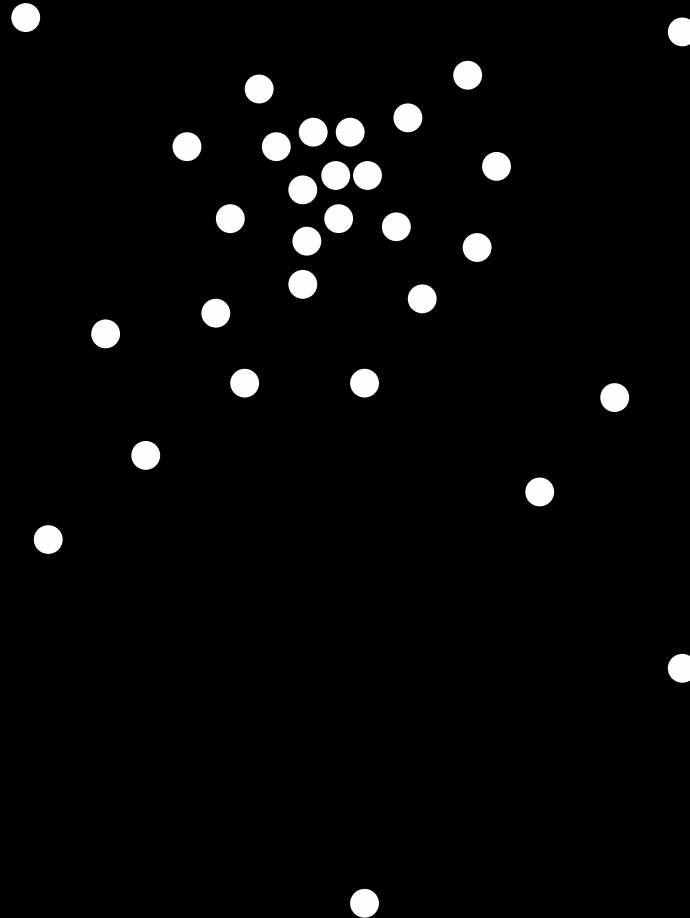
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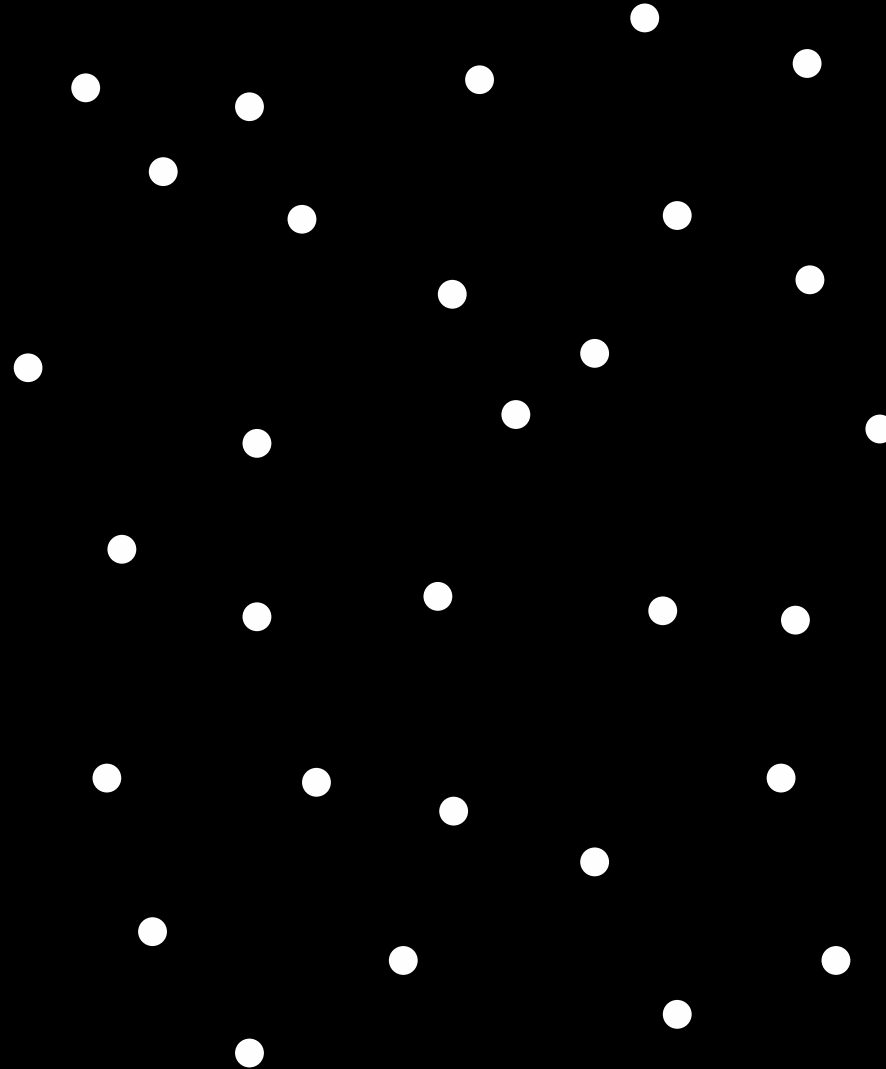
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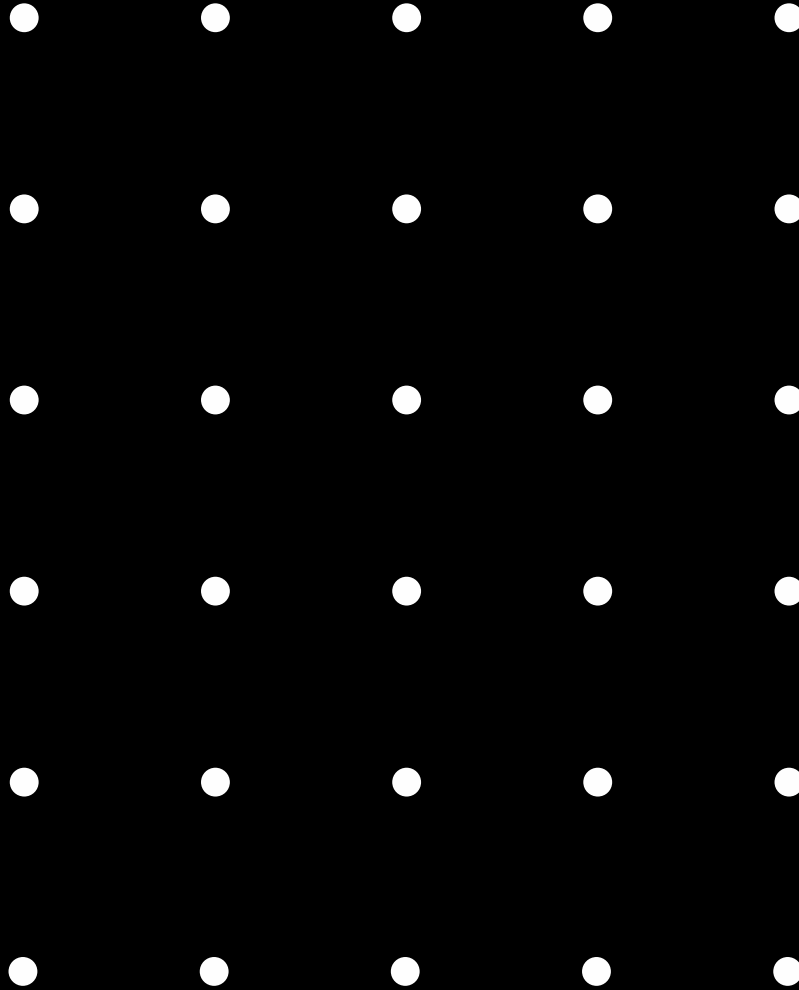
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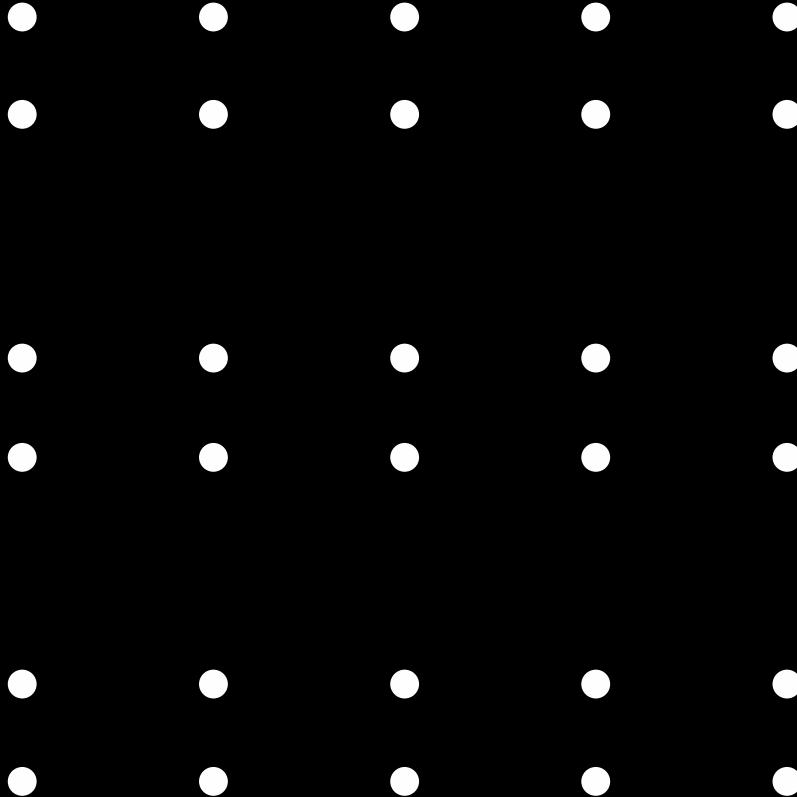
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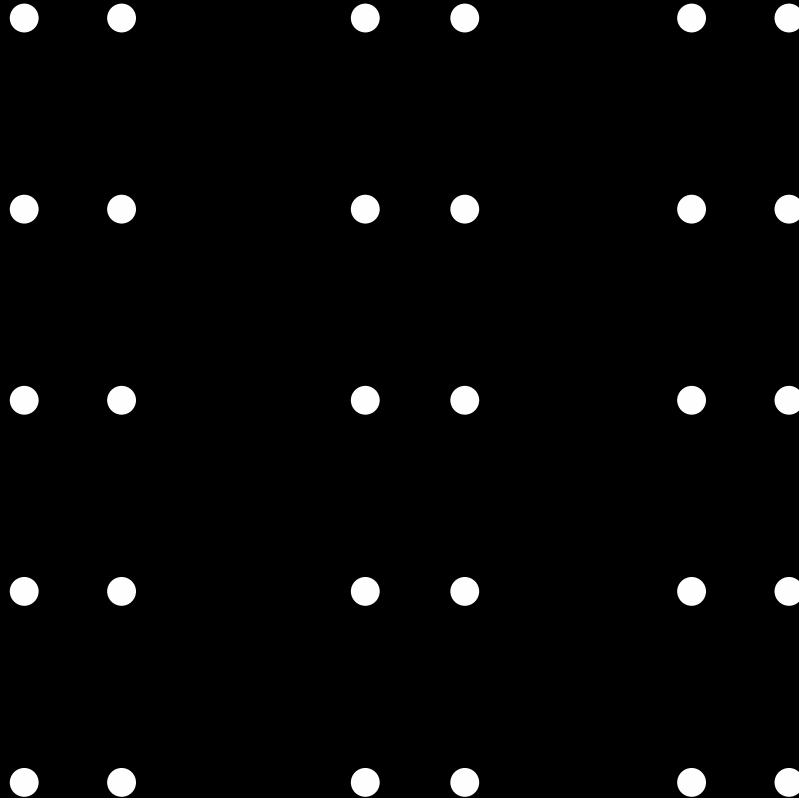
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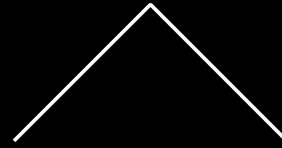
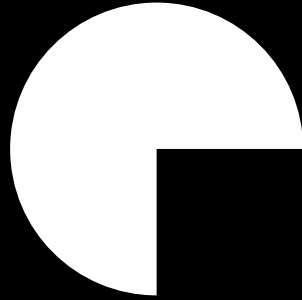
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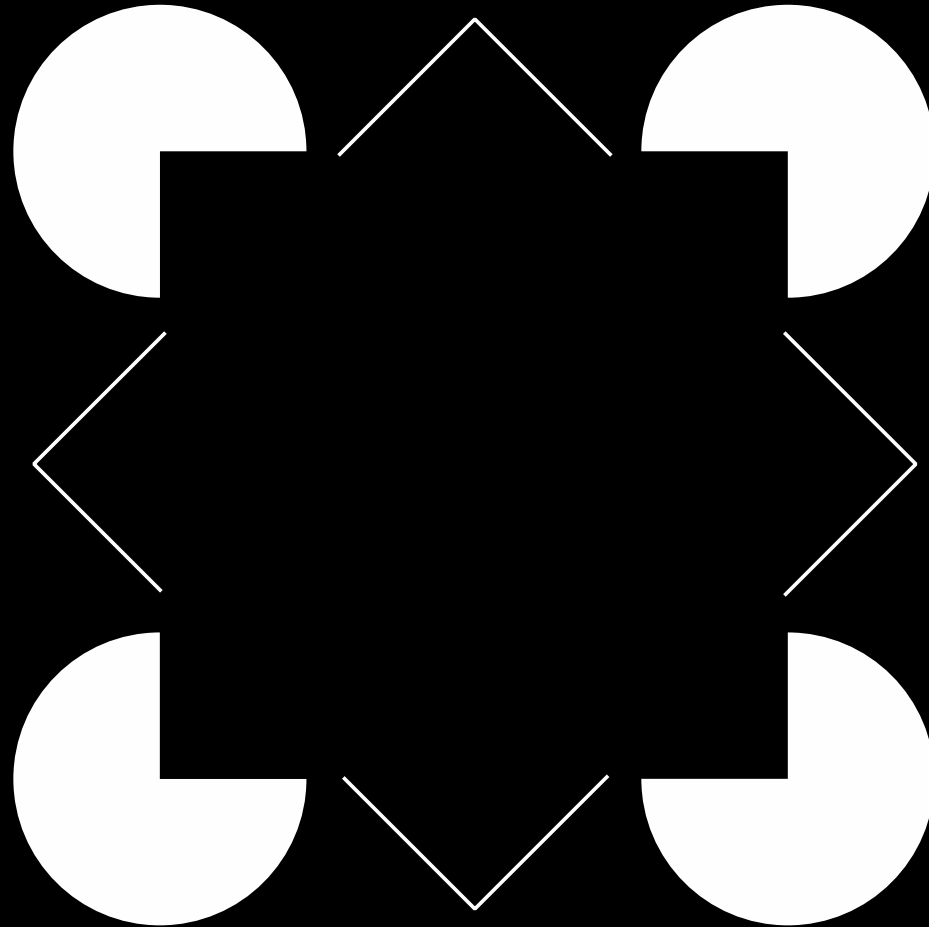
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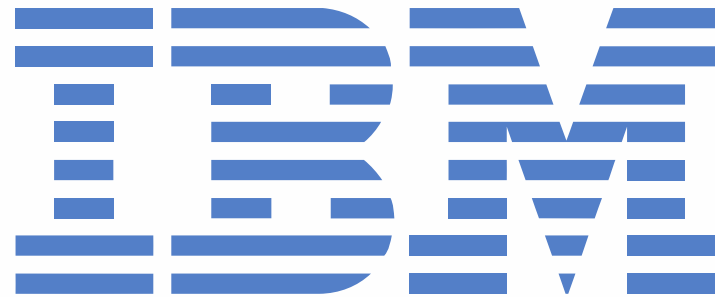
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The word "CONTINUATION" is written in a bold, white, sans-serif font on a dark gray rectangular background. The letters are closely spaced. The 'O' in "CONTINUATION" is stylized with a horizontal line passing through its center, which extends to the right, ending in an arrowhead. This visual element demonstrates the principle of continuation by showing how the eye follows the path of the line through the letters.

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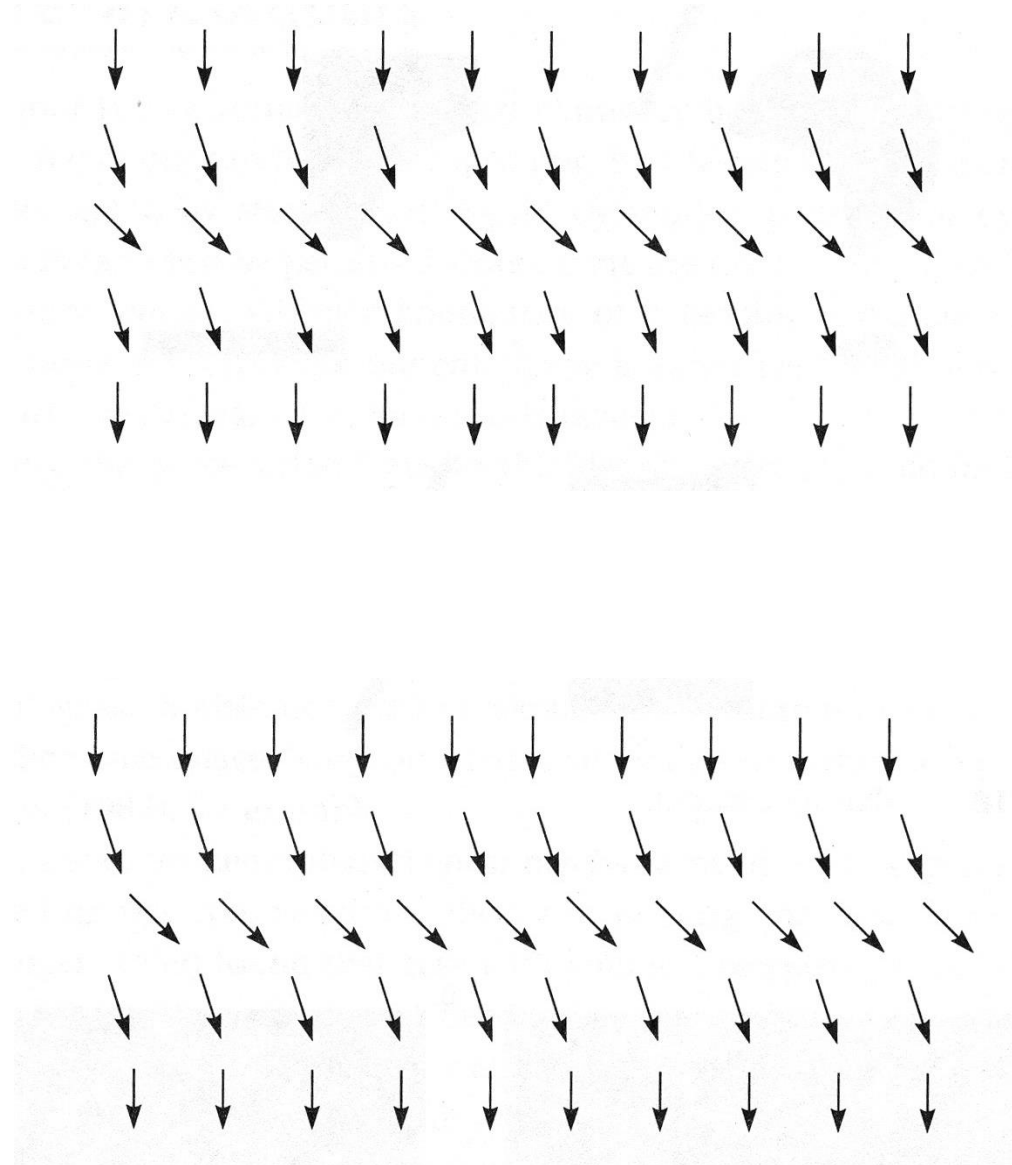
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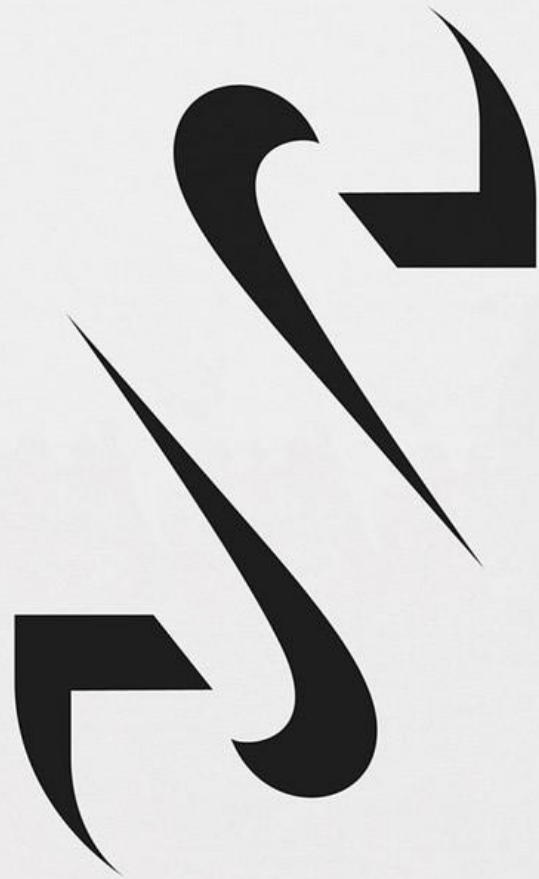
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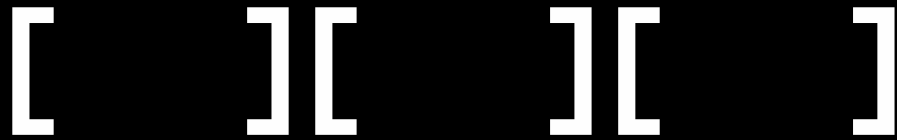
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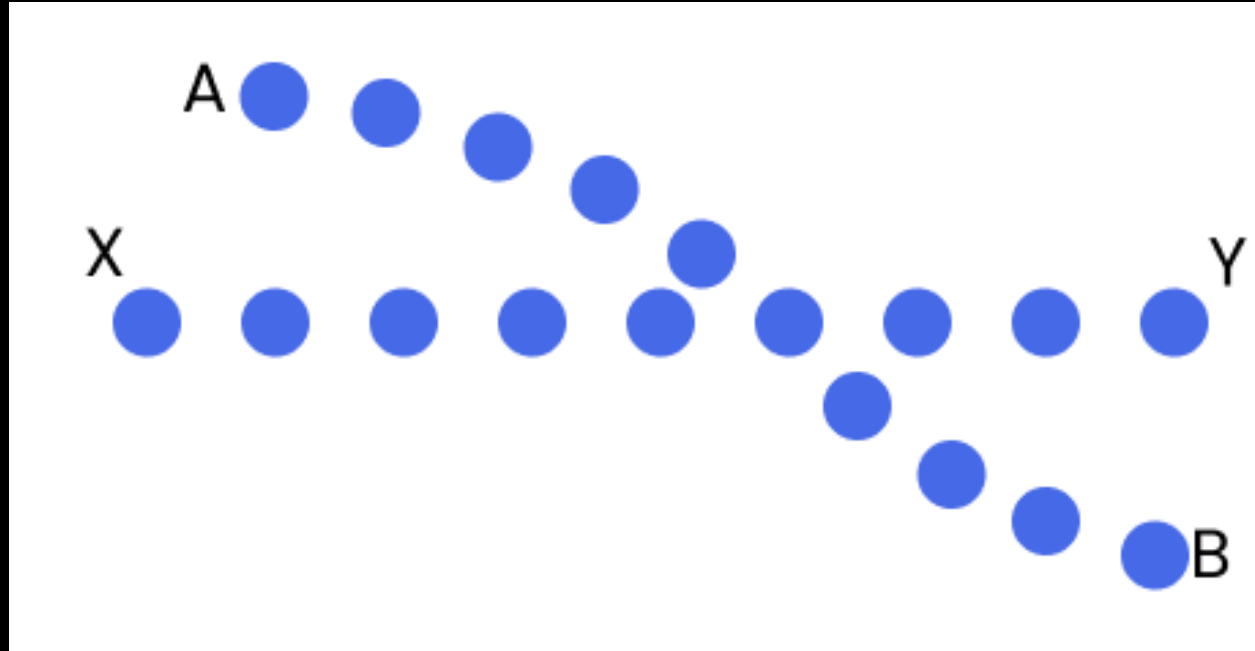
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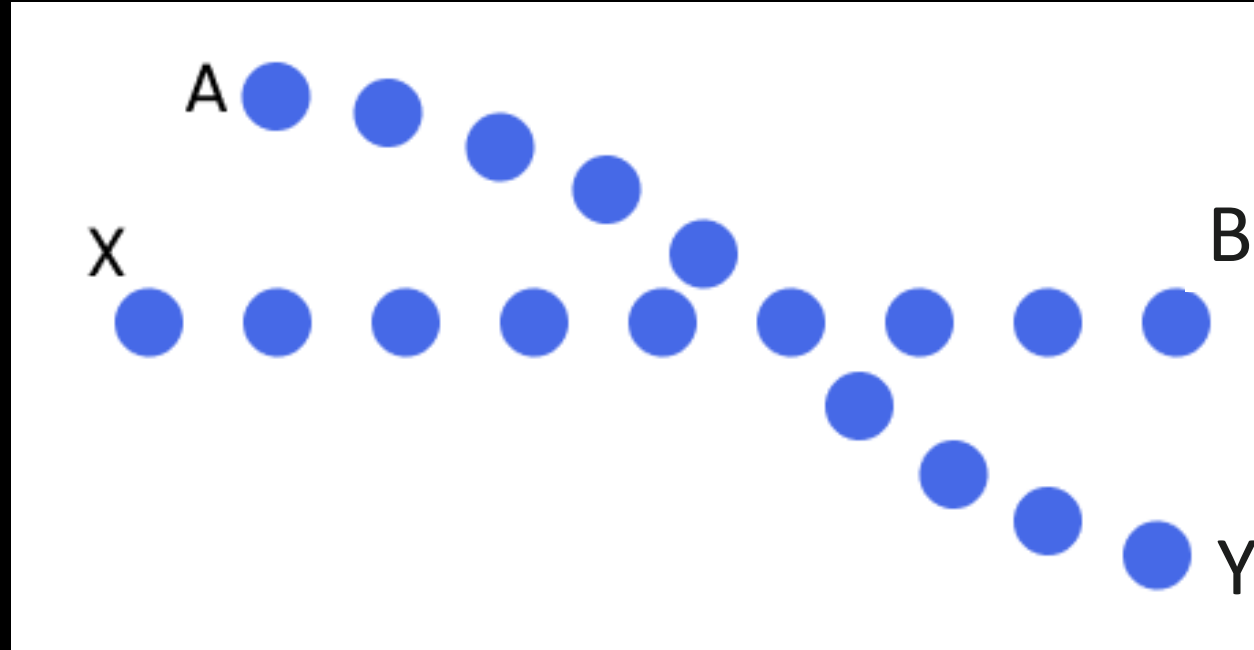
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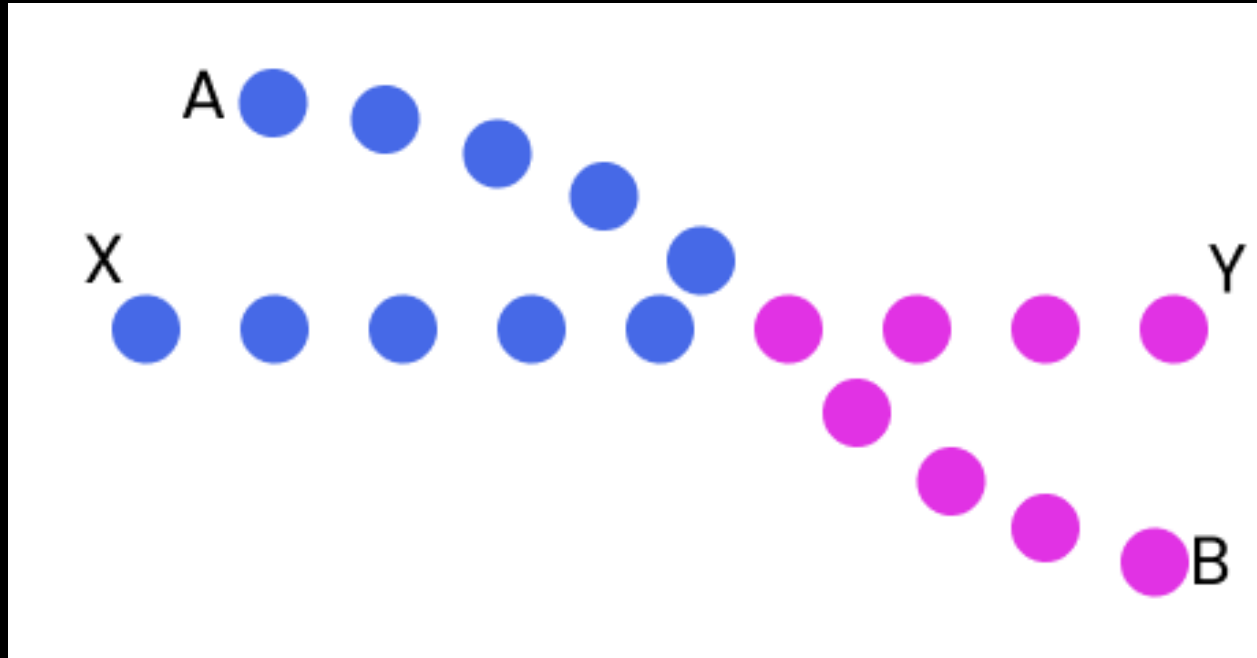
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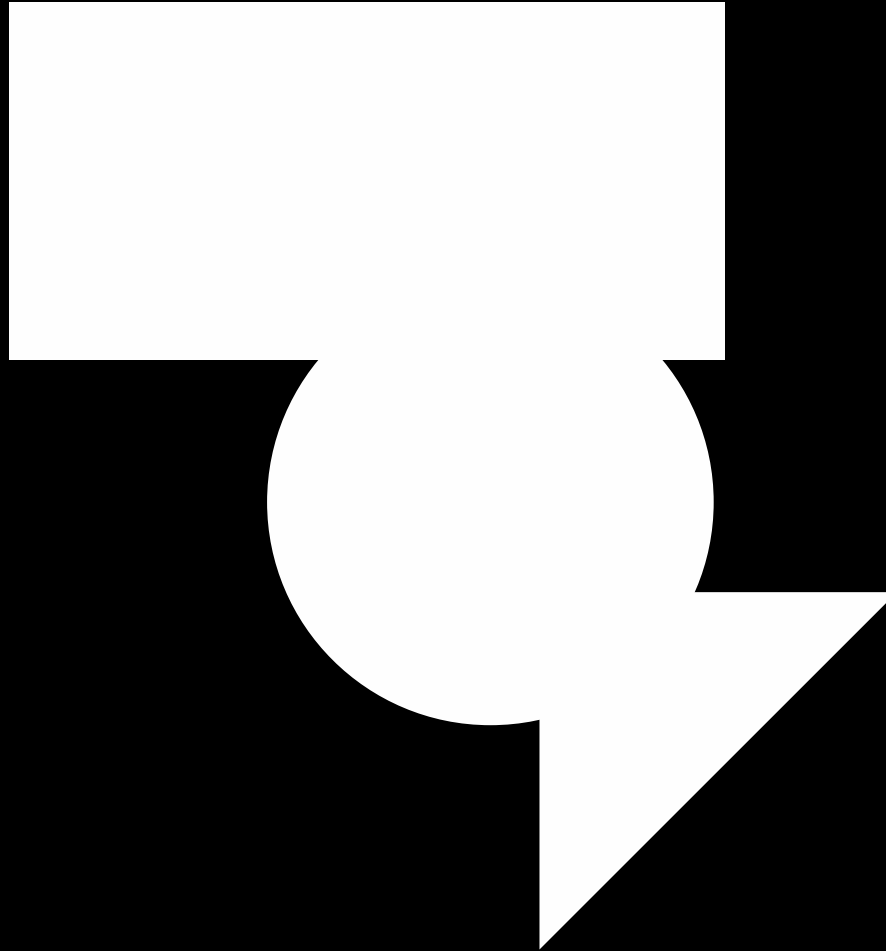
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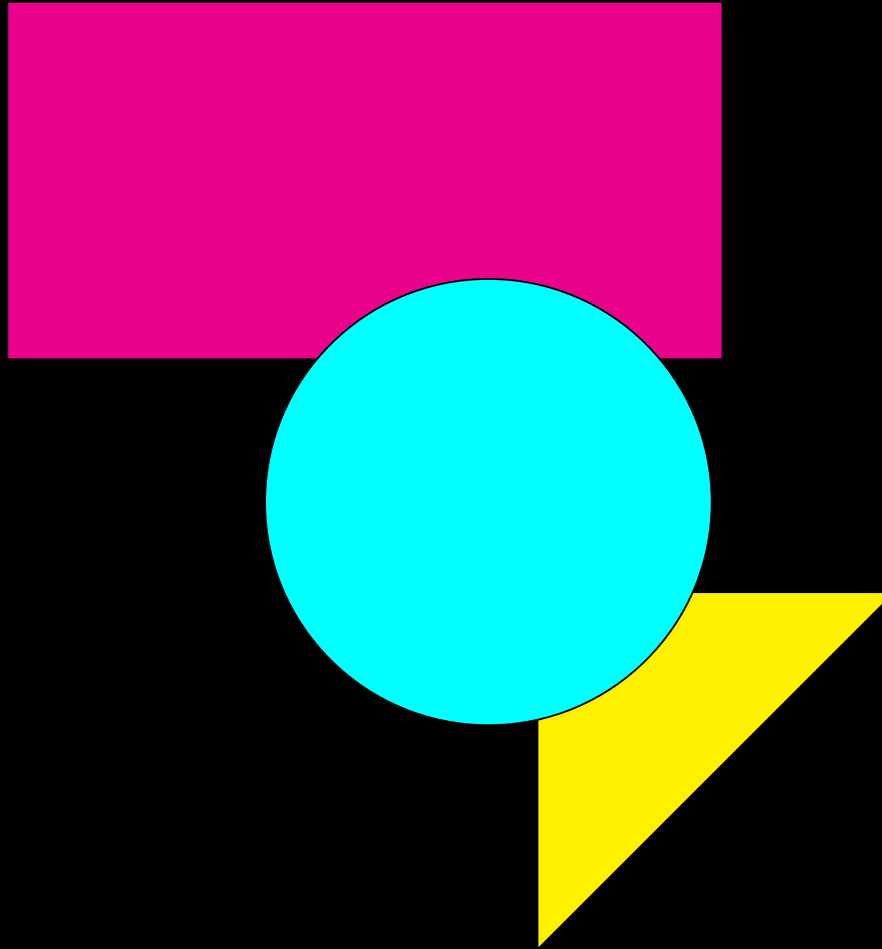
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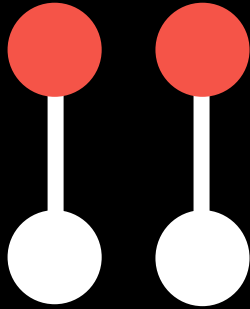


Bonus!

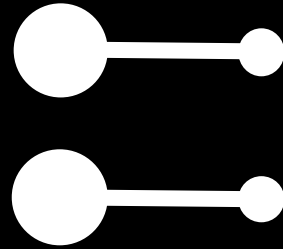
Connectedness



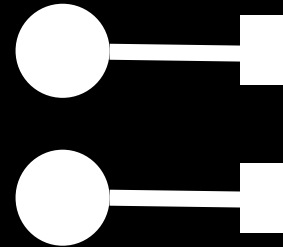
proximity



color



size



shape

Change Blindness

An interruption in what is being seen causes us to miss significant changes that occur in the scene during the interruption.



Selective Attention

Selective attention is the filtering of information to focus on the most relevant information



Selective Attention

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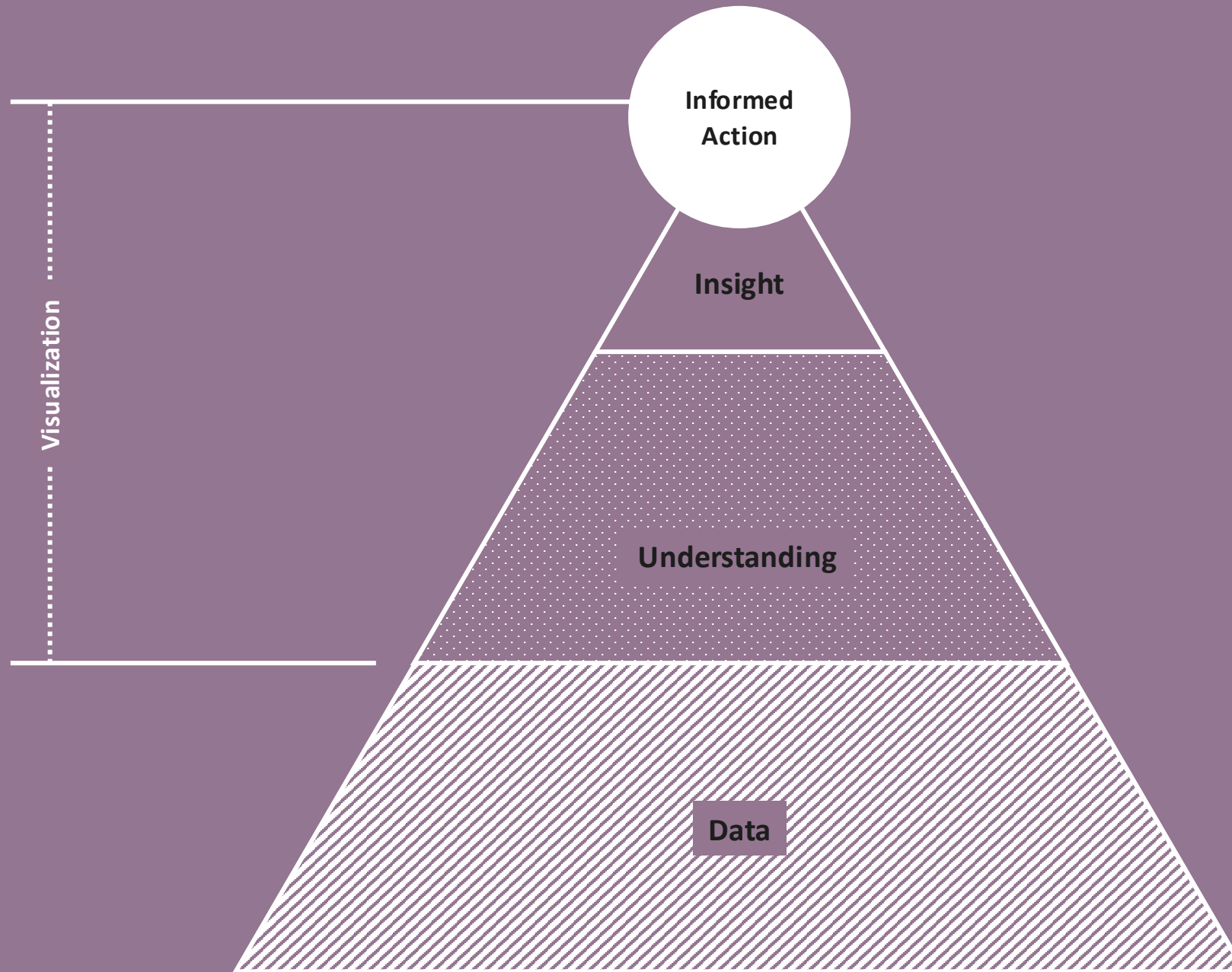


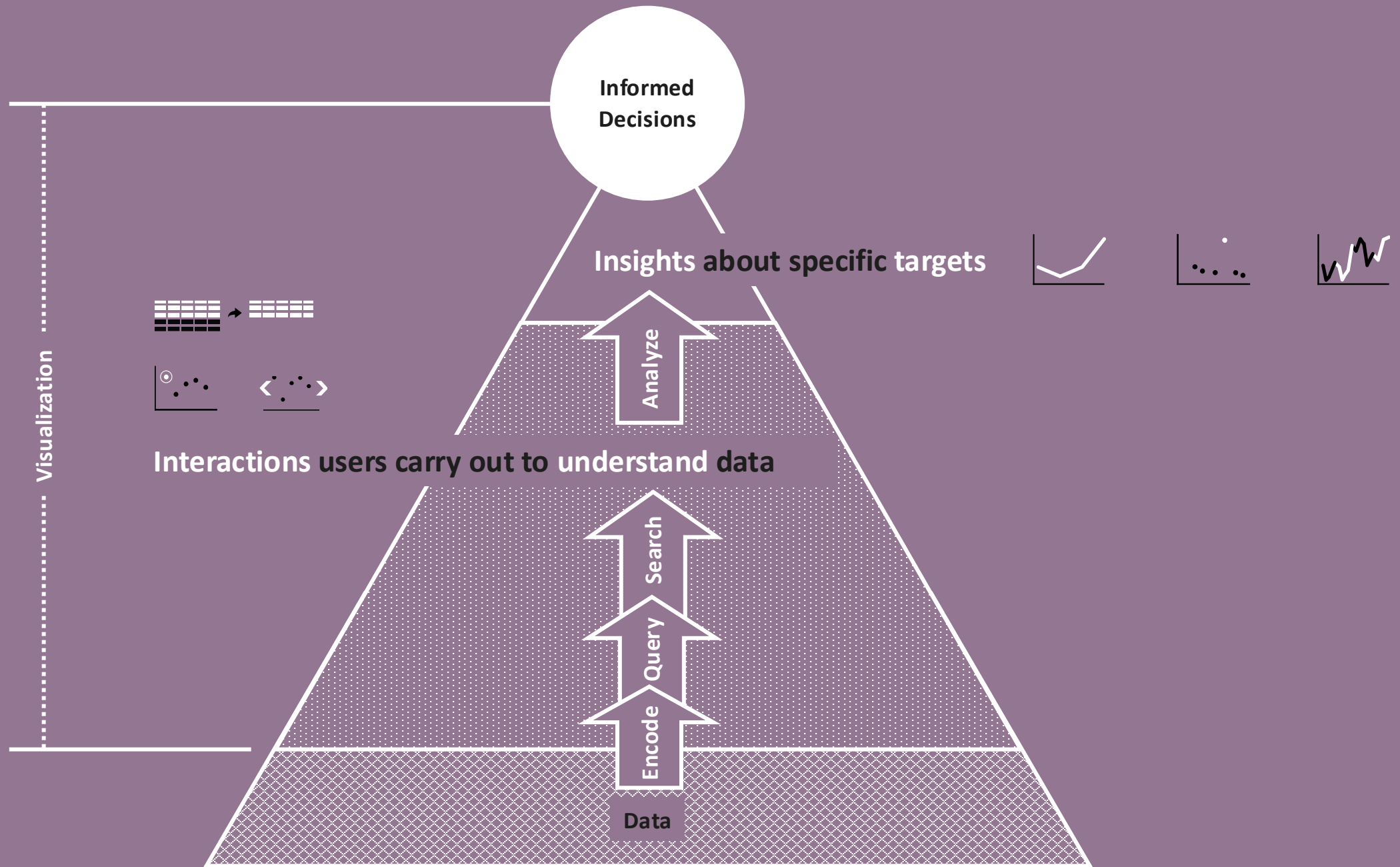
Break
until 7:15 pm

Tasks & Interactions

**“The use of computer-
supported, interactive,
visual representations of
abstract data to amplify
cognition”**

Card, MacKinlay, and Shneiderman, 1999





Increasing human **memory and processing** resources

Reducing **search** time for information

Enhancing **pattern detection** based on human perception

Enabling **inference** operations (e.g., by making problems obvious)

Supporting **perceptual monitoring** and controlling user attention

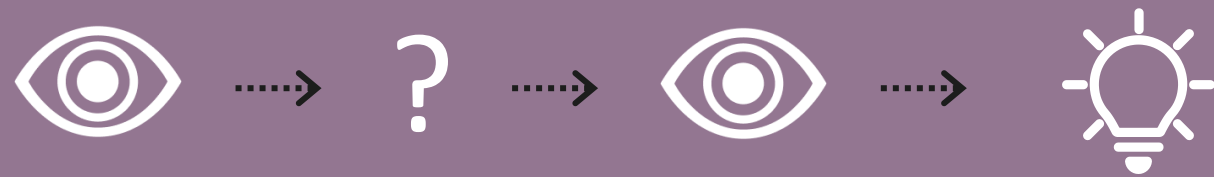
Encoding data in a **manipulable** medium, to improve cognition

Tasks

Directed



Explorative



Tasks

- 1** How much did this particular variety of shirt cost to produce in week 3?
- 2** Which weeks in March did we produce 100 shirts?
- 3** Is production rising or falling for any particular clothing line?
- 4** Are dresses keeping up with shirts?

- A** See patterns, trends, outliers
- B** Compare attributes of two objects
- C** Find value of specific target
- D** Find target when we have a value

Tasks

1 How much did this particular variety of shirt cost to produce in week 3?

2 Which weeks in March did we produce 100 shirts?

3 Is production rising or falling for any particular clothing line?

4 Are dresses keeping up with shirts?

A See patterns, trends, outliers

B Compare attributes of two objects

C Find value of specific target

D Find target when we have a value

Tasks

Actions

Analyze

Search

Query

Targets

All data

Attributes

Network data

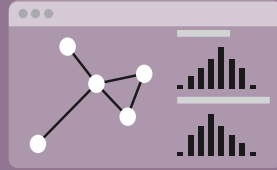
Spatial data

Actions: Analyze

Consume



Discover

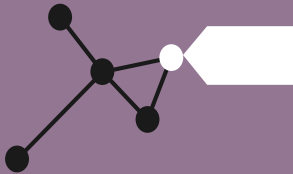


Present

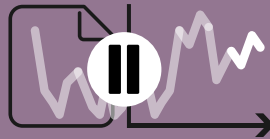


Enjoy

Produce



Annotate



Record



Derive

Actions: Search

Location
known

Location
unknown

Target
known



Lookup



Locate

Target
unknown

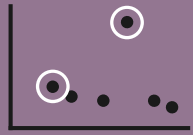


Browse

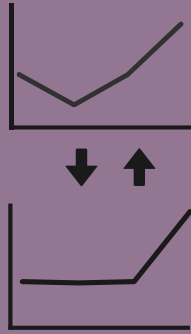


Explore

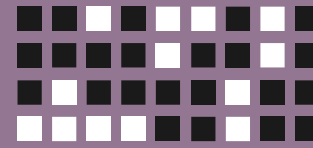
Actions: Query



Identify



Compare

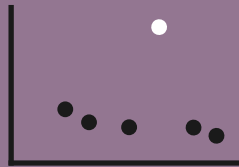


Summarize

Targets: All Data



Trends



Outliers



Features

Targets: Attributes

One



Distribution

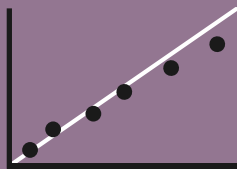


Extremes

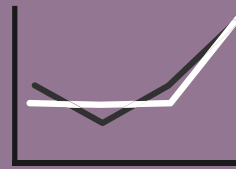
Many



Dependency



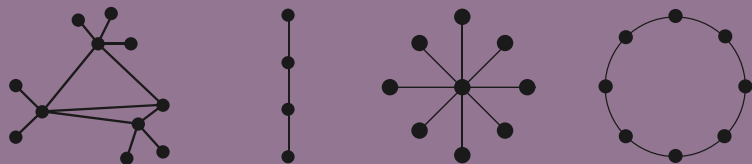
Correlation



Similarity

Targets: Networks

Targets: All Data



Topology

Targets: Attributes



Paths /
lines



Points /
nodes

SOCIAL POWER GUIDE

Power via: **Love/Family** Coalitions **Leaders** Connections

Li Yuanchao

Mentor to

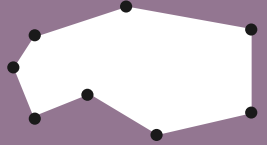
Hu Jintao 胡锦涛
China's sixth president, former Party general secretary and former chairman of the Central Military Commission

Profile Inst'l Compare

- Hu Jingzhi *Father*
- Li Wenrui *Mother*
- Liu Yongqing *Wife*
 - Hu Haiqing *Daughter*
 - Hu Haifeng *Son*
- Hu Jinrong *Younger sister*
- Hu Jinlai *Sister*

First Degree Second Degree

Targets: Spatial



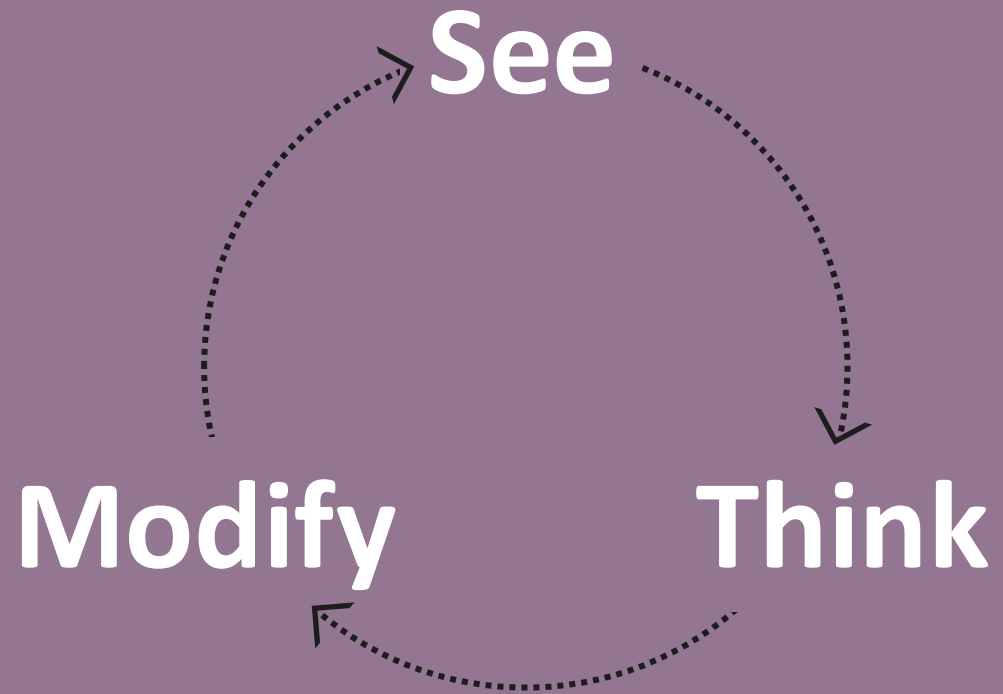
Shape

Activity

In your project groups

- 1 Find an example of a “good” visualization online. Analyze an Action that it supports and its Target(s).**
- 2 Find another supported Action with related Targets.**
- 3 Make quick notes to explain your analysis.**
- 4 Add a screenshot of the visualization in a Google Slide**
- 5 Present your example to the class**

20 minutes



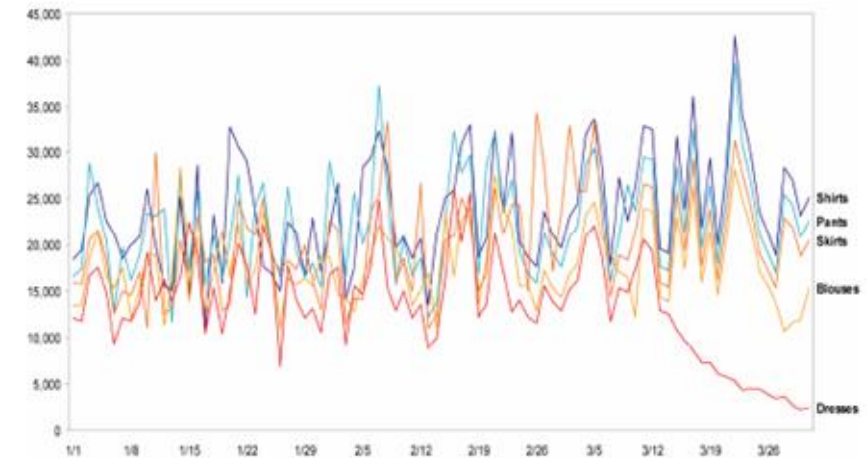
Inter Action

**“Overview first,
zoom and filter, then
details-on-demand”**

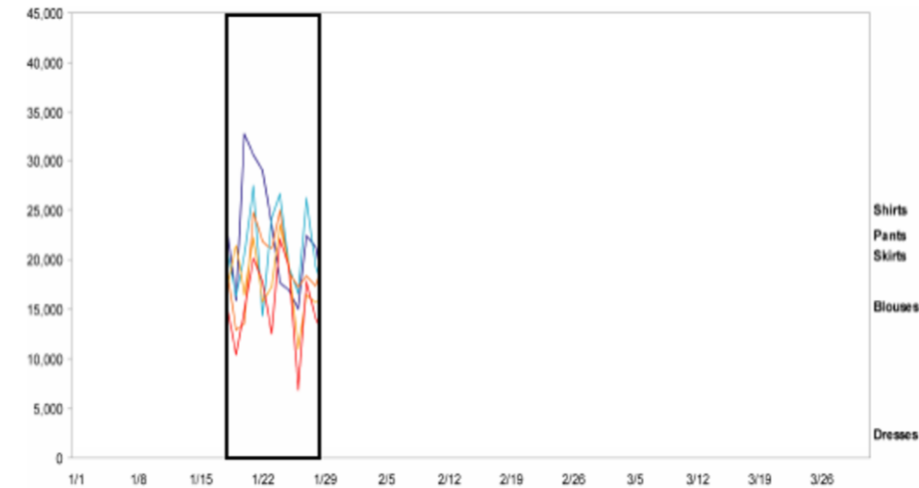
Shneiderman, 1996

Actions

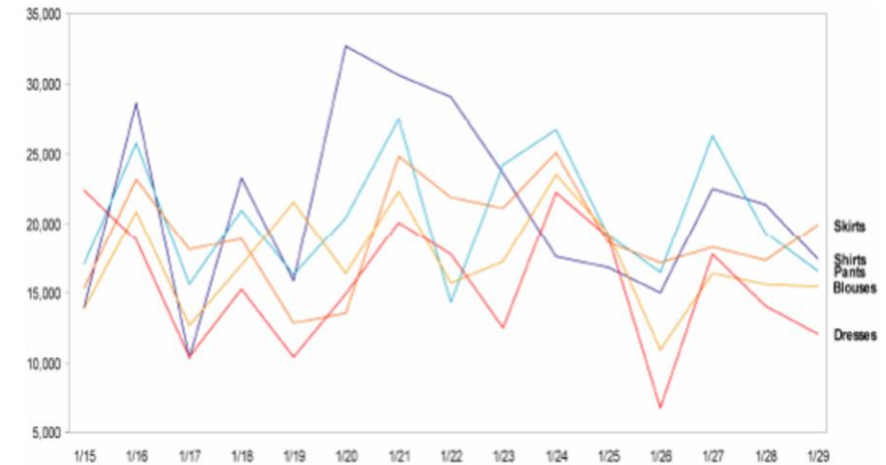
“Overview first,
zoom and
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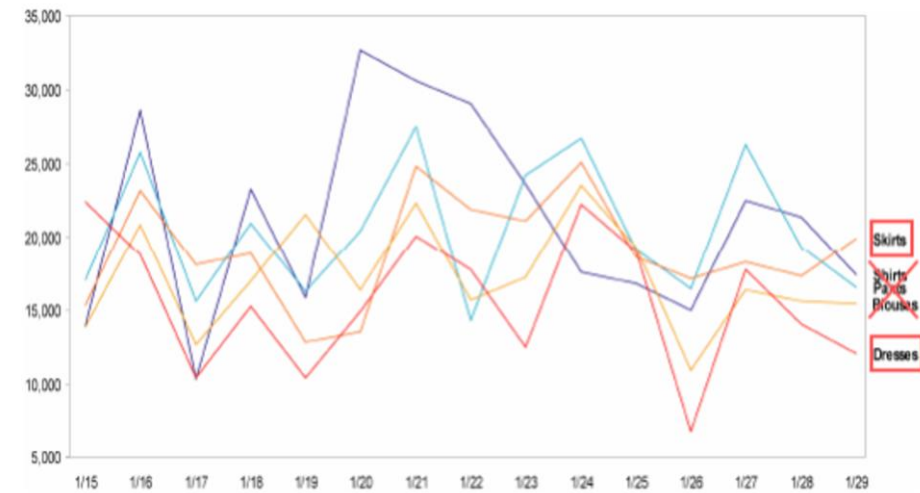
“Overview first,
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“Overview first,
zoom and
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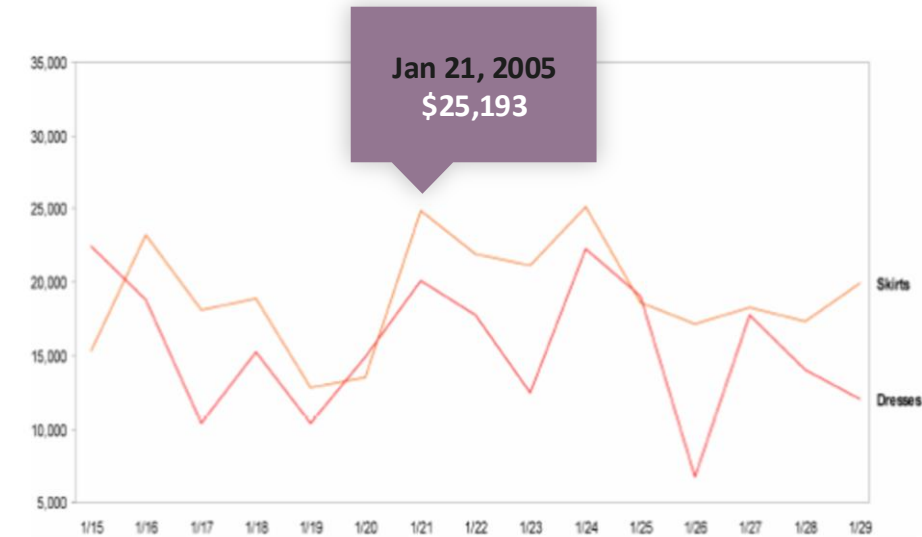
“Overview first,
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“Overview first,
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filter, then
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“Overview first,
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“Overview first,
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relate

see relationships, compare values

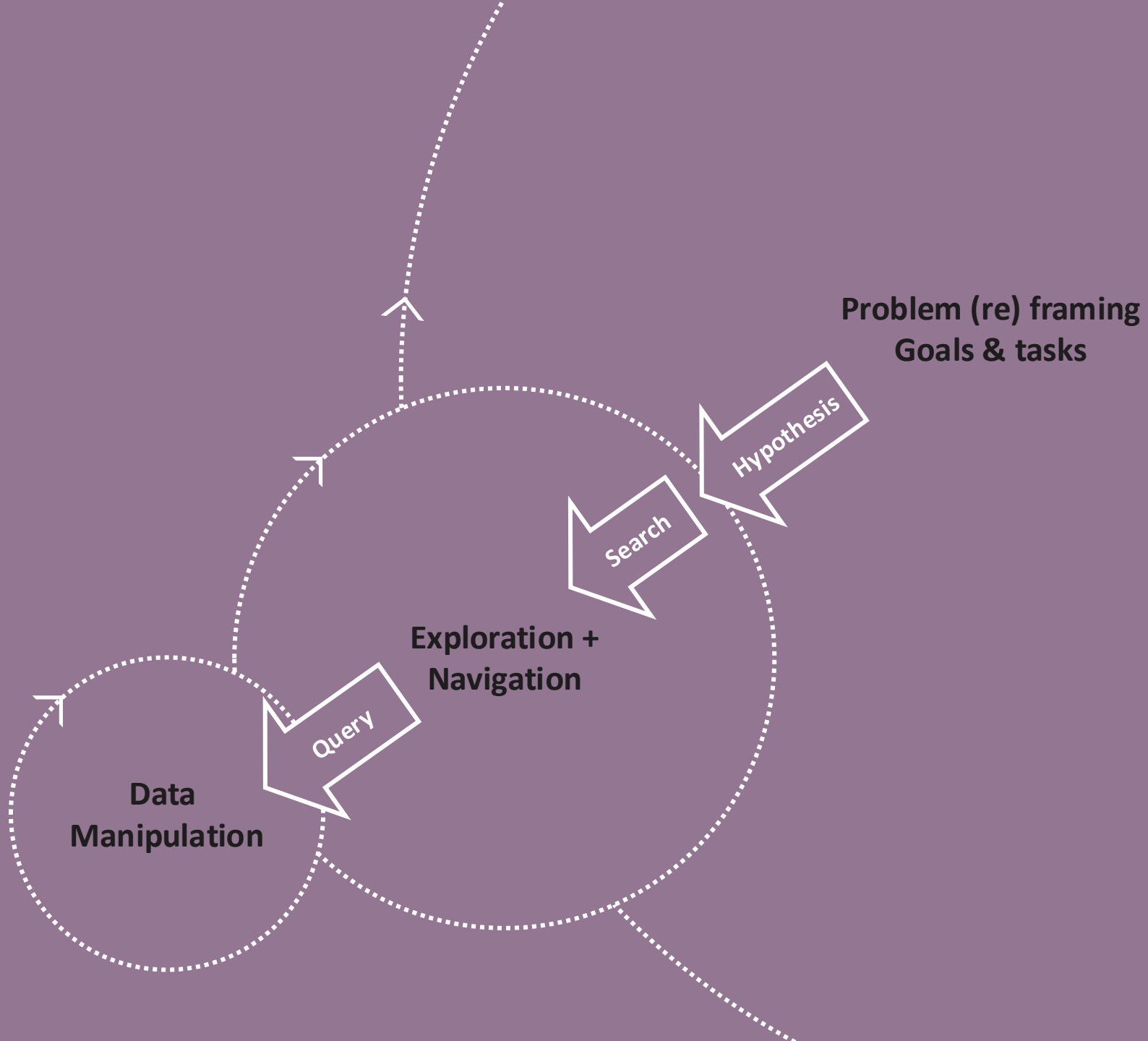
history

keep track of actions and insights

extract

mark and capture data

Interaction Loops



Data Specification

Visualize

Filter

Sort

Derive

Aggregate

Separate

...

View Manipulation

Select

Navigate

Coordinate

Organize

Facet (juxtapose,
partition, superimpose)

Embed

...

Advanced

Brushing and Linking
(Highlighting)

Dynamic Queries (Filtering)

Overview + Detail

Focus + Context

Panning and Zooming

Distortion-based Views

...

Data Specification

Visualize

Filter

Sort

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View Manipulation

Select

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Advanced

**Brushing and Linking
(Highlighting)**

Dynamic Queries (Filtering)

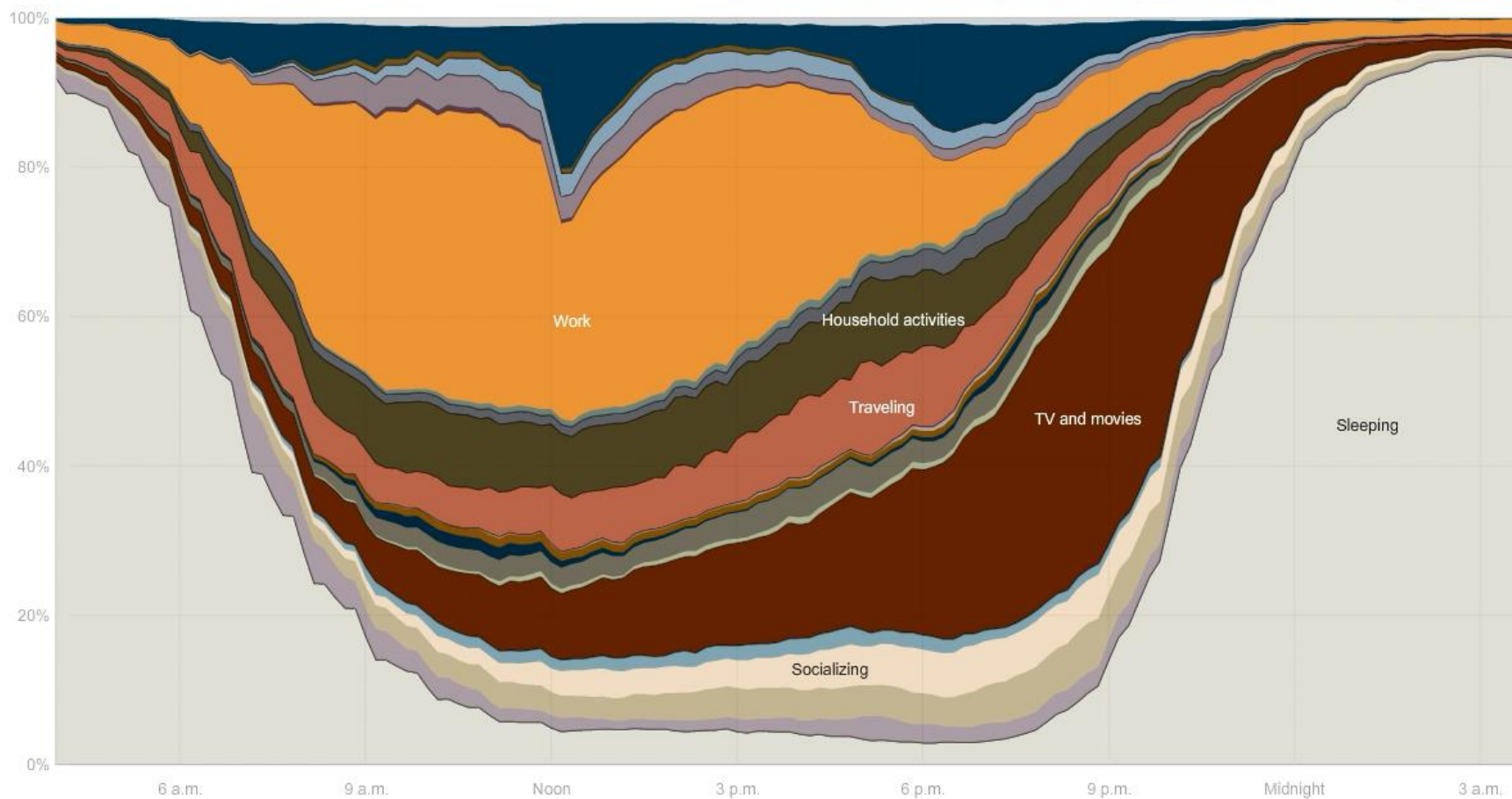
Overview + Detail

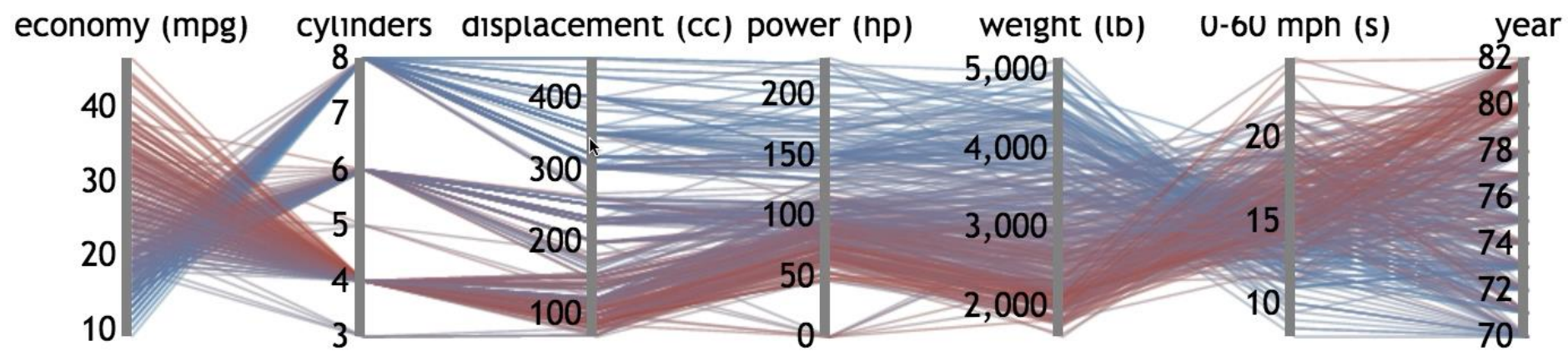
Focus + Context

Panning and Zooming

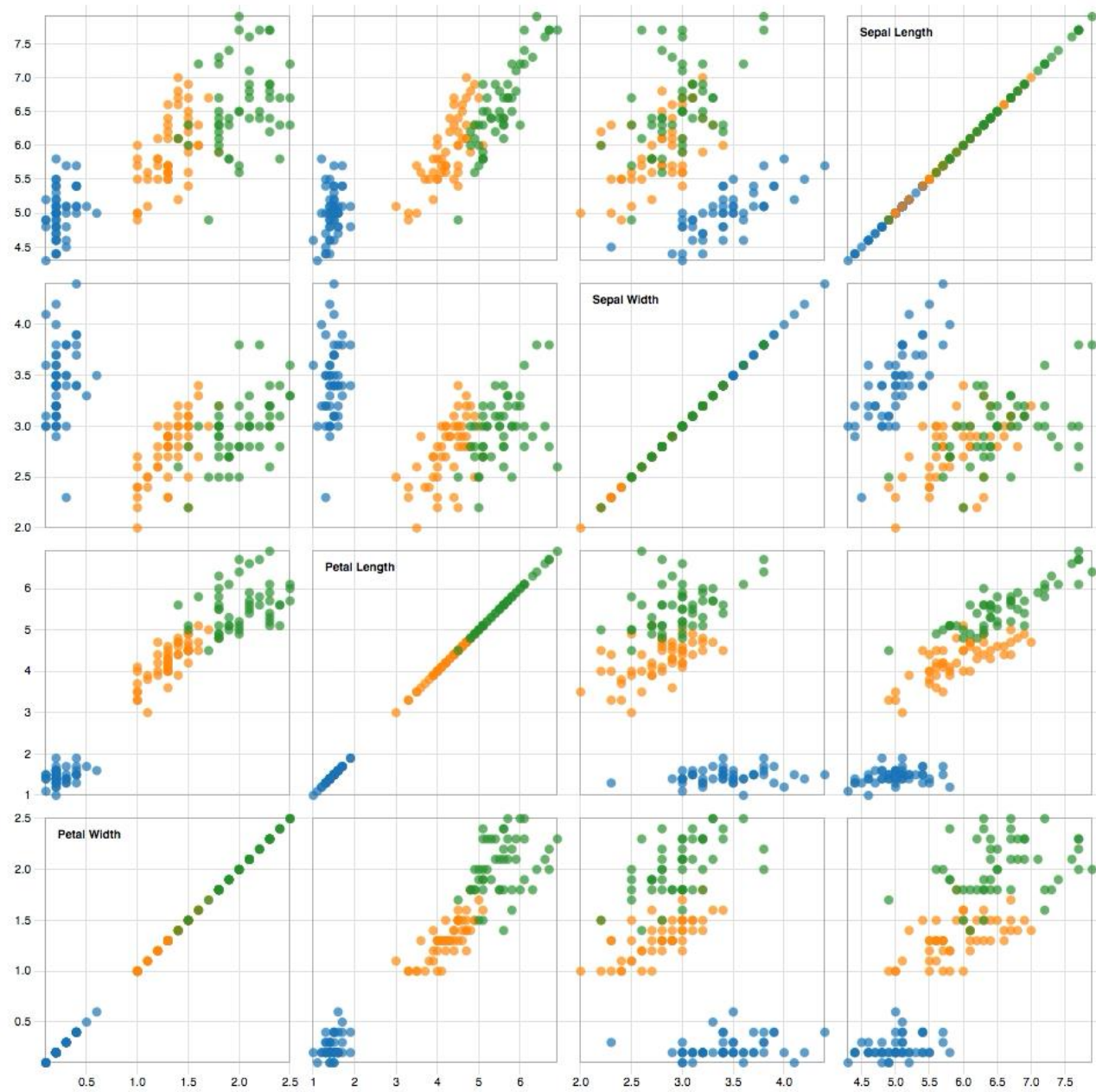
Distortion-based Views

...

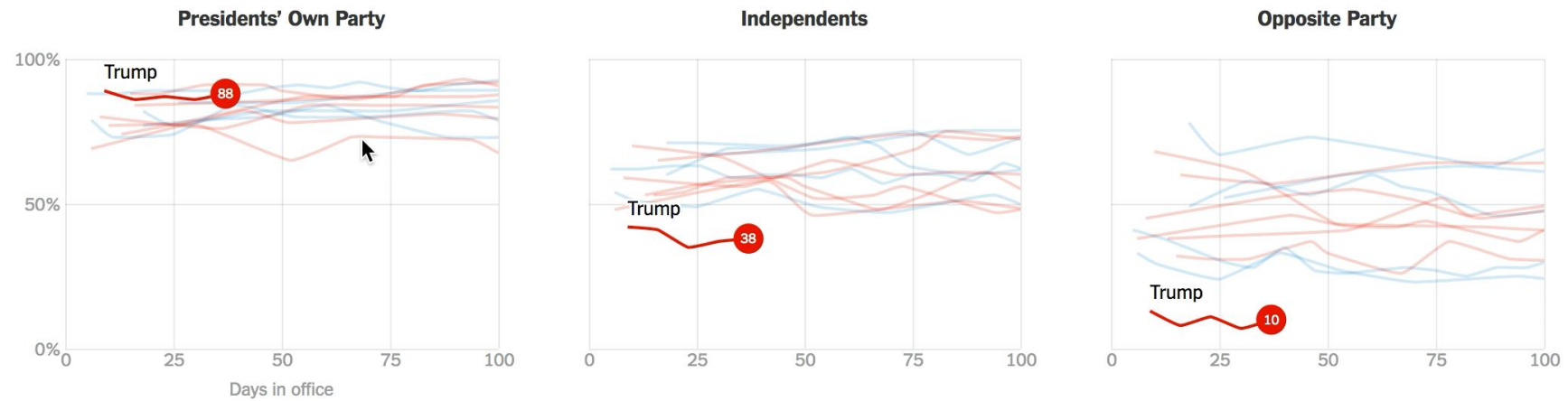




Brushing + Linking



How Trump's Approval Rating Compares To Previous Presidents

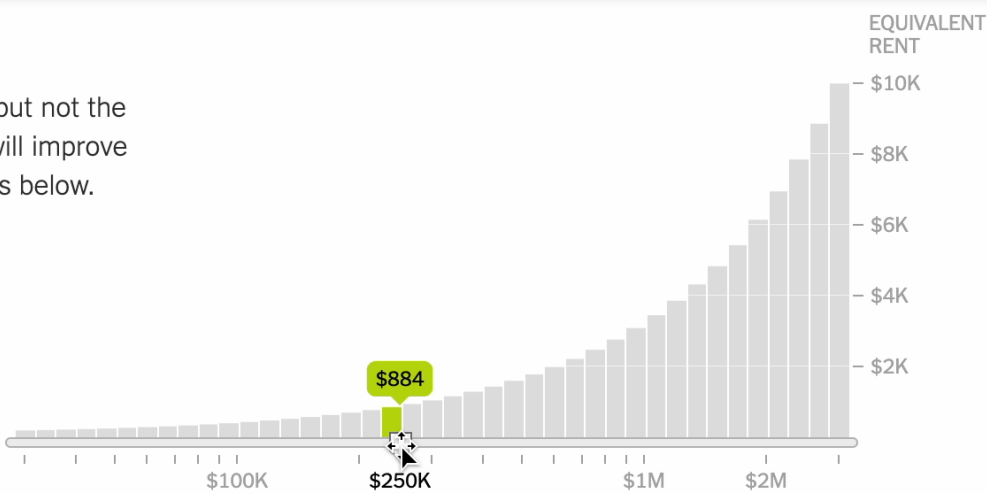




Home Price

A very important factor, but not the only one. Our estimate will improve as you enter more details below.

\$250,000



If you can rent a similar home for less than ...

\$884 PER MONTH

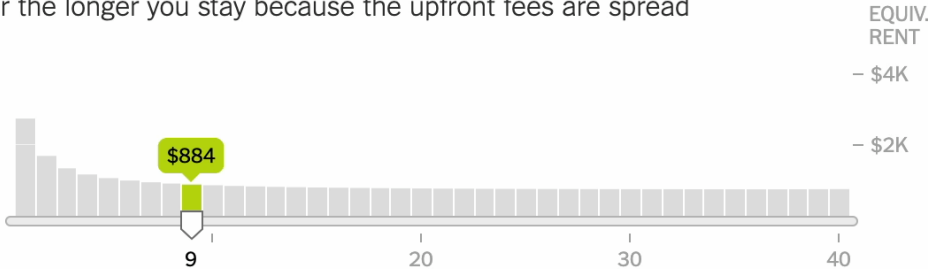
... then renting is better.

Costs after 9 years	Rent	Buy
Initial costs	\$884	\$60,000
Recurring costs	\$106,941	\$163,398
Opportunity costs	\$15,396	\$44,587
Net proceeds	-\$884	-\$145,649
Total	\$122,337	\$122,337

How Long Do You Plan to Stay?

Buying tends to be better the longer you stay because the upfront fees are spread out over many years.

9 years



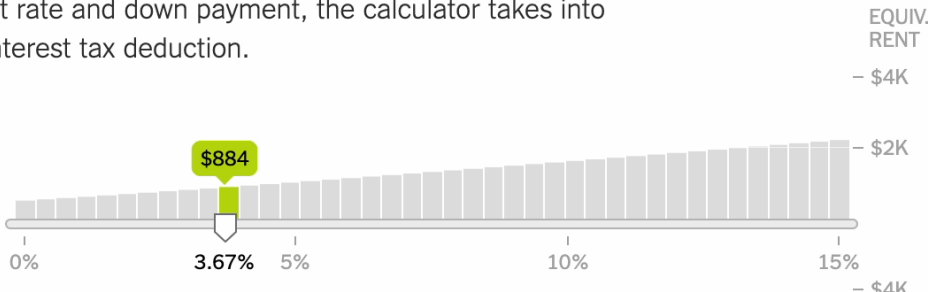
How to Read the Charts Charts that are relatively flat indicate factors that are not particularly important to the outcome. Conversely, the factors that have steep slopes have a large impact.

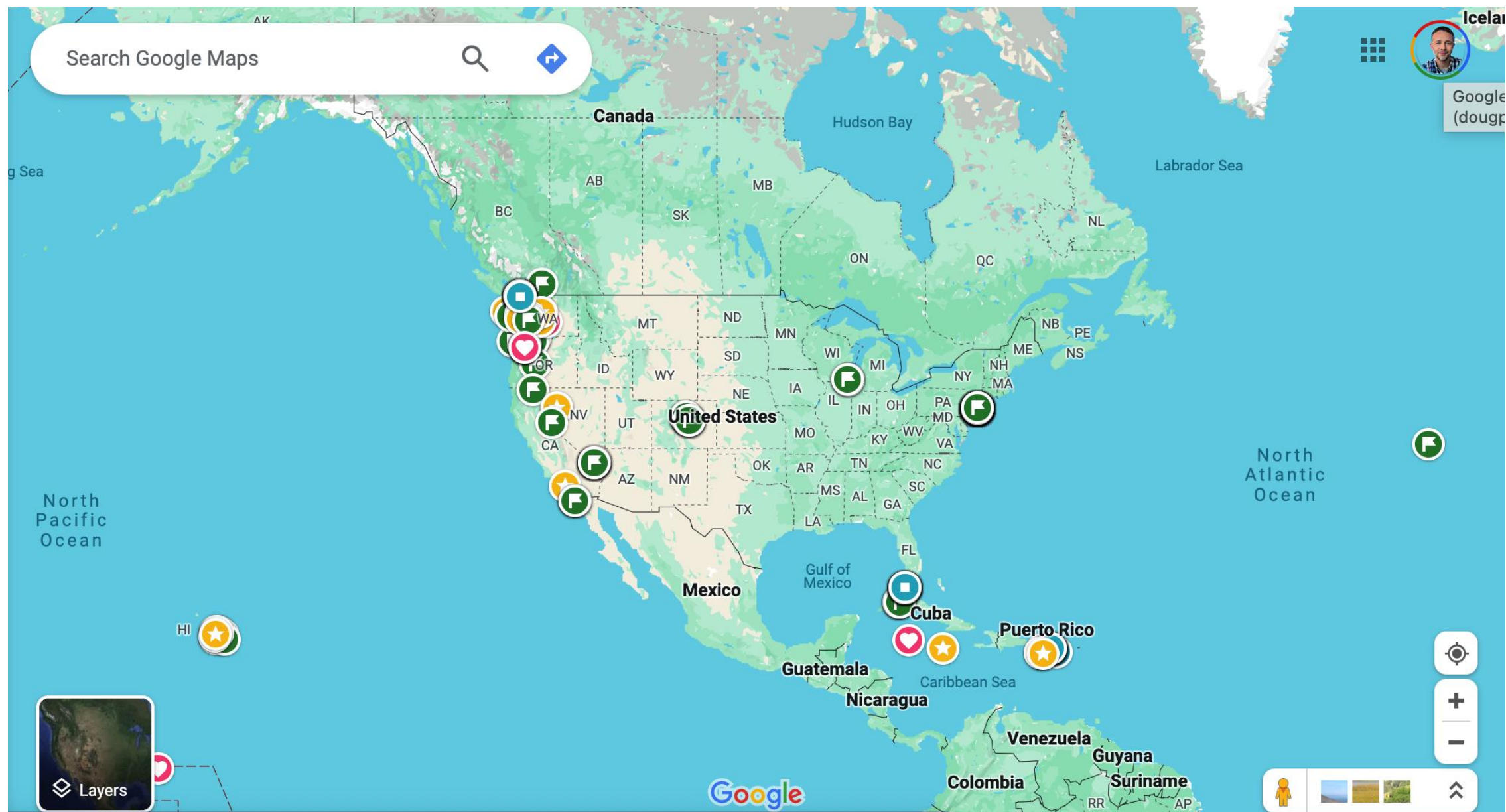
What Are Your Mortgage Details?

In addition to the interest rate and down payment, the calculator takes into account the mortgage-interest tax deduction.

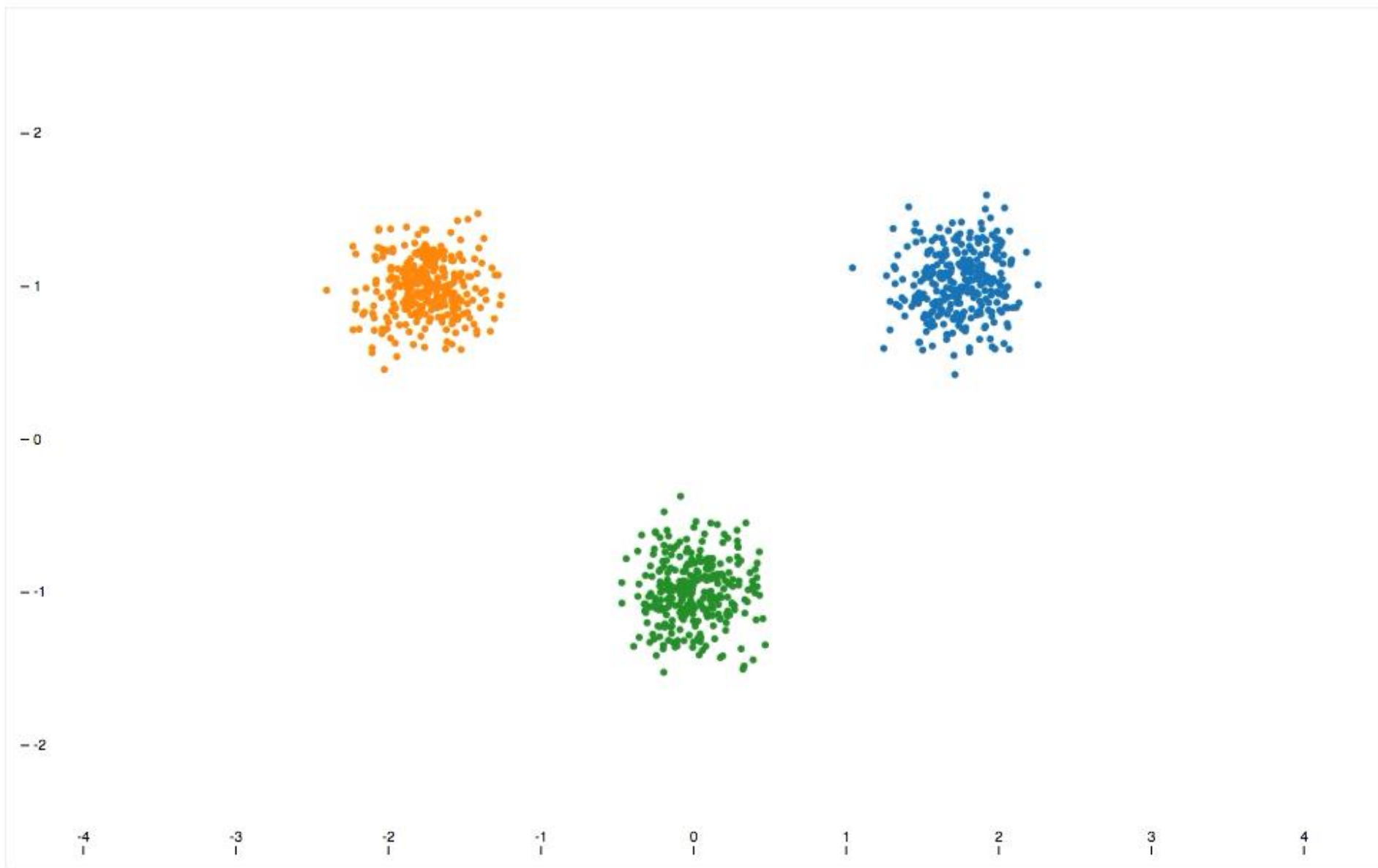
3.67%

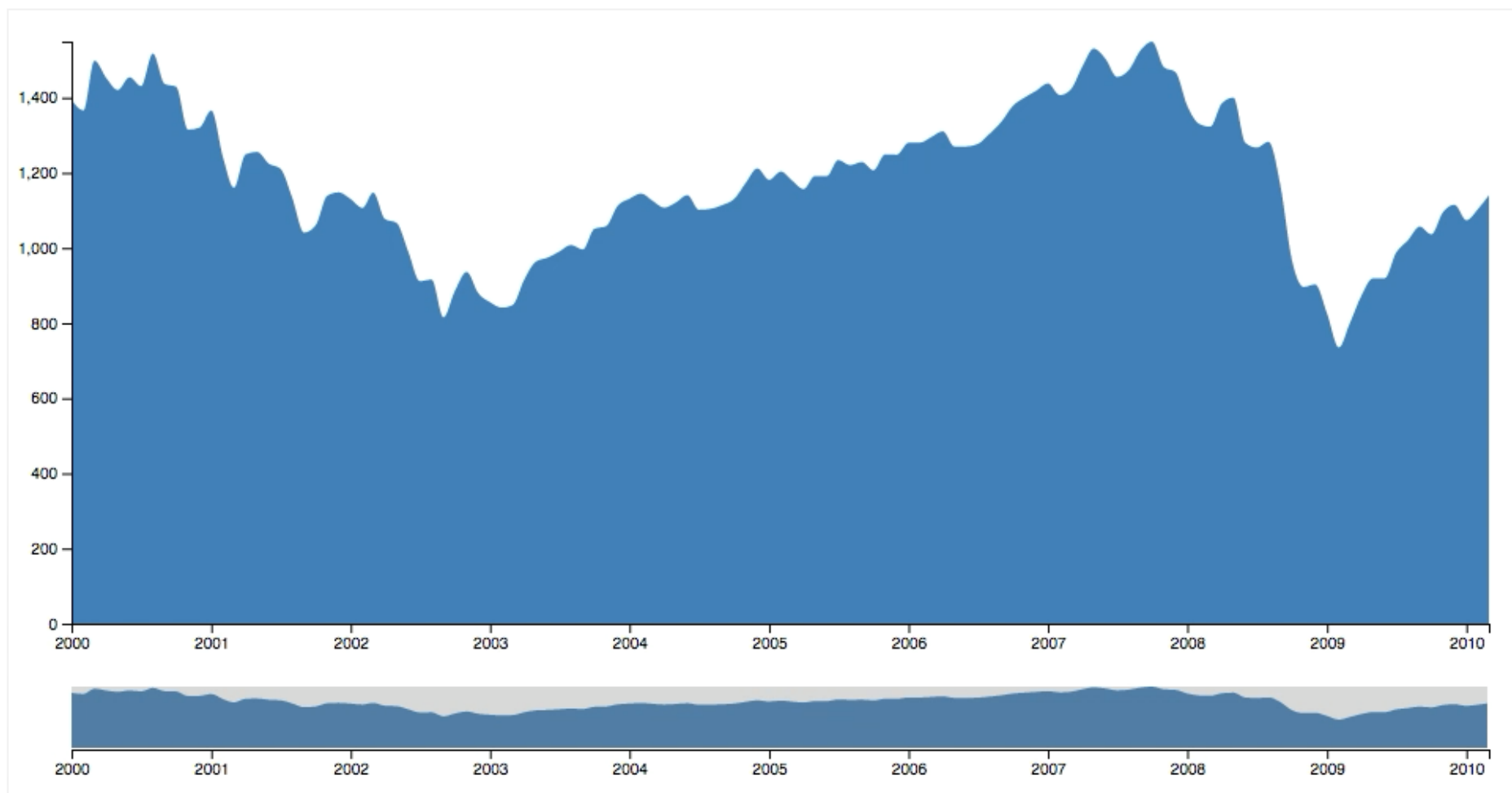
Mortgage rate ?
\$917 per month

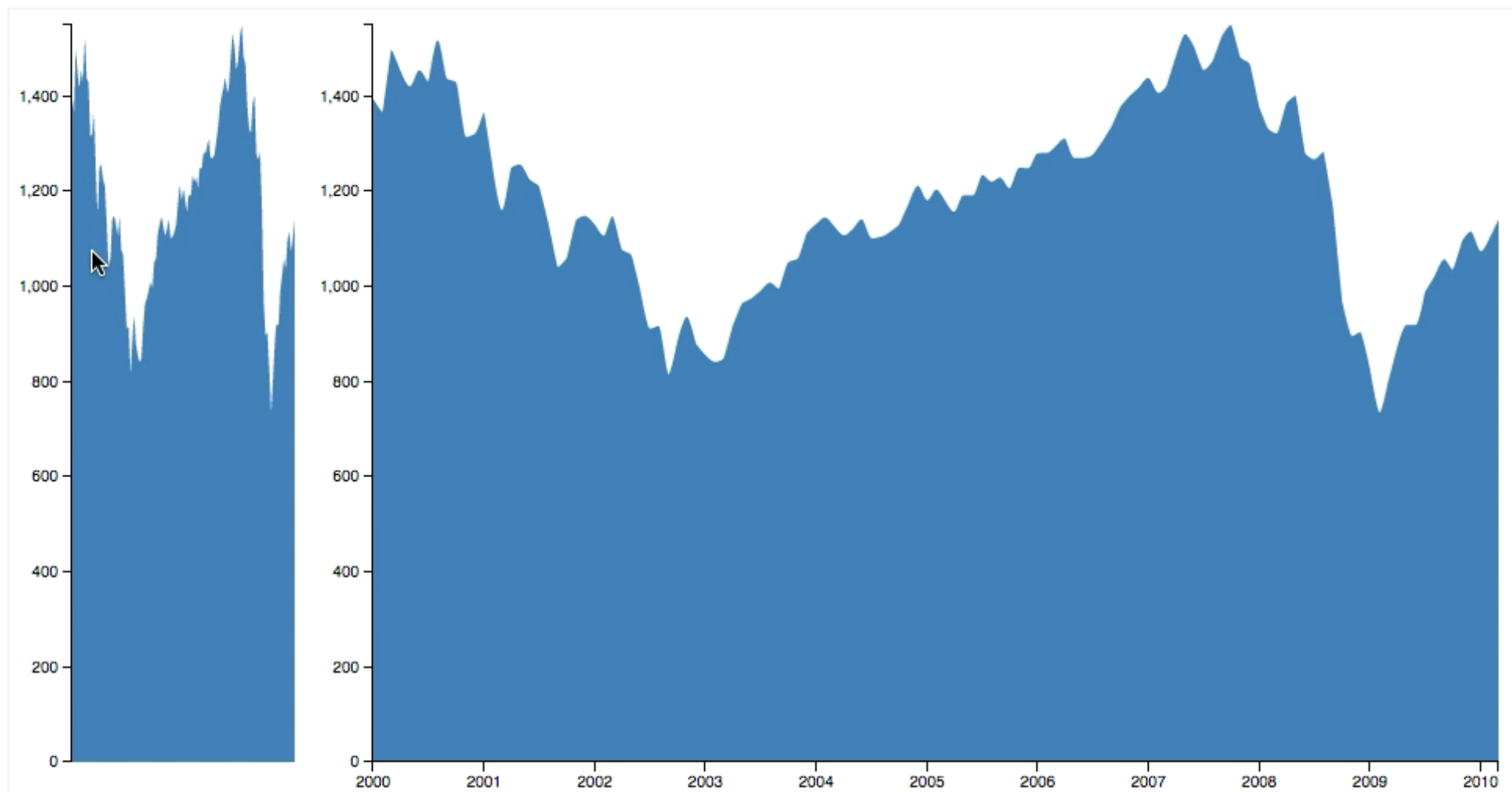




Focus + Context







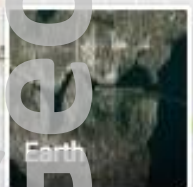
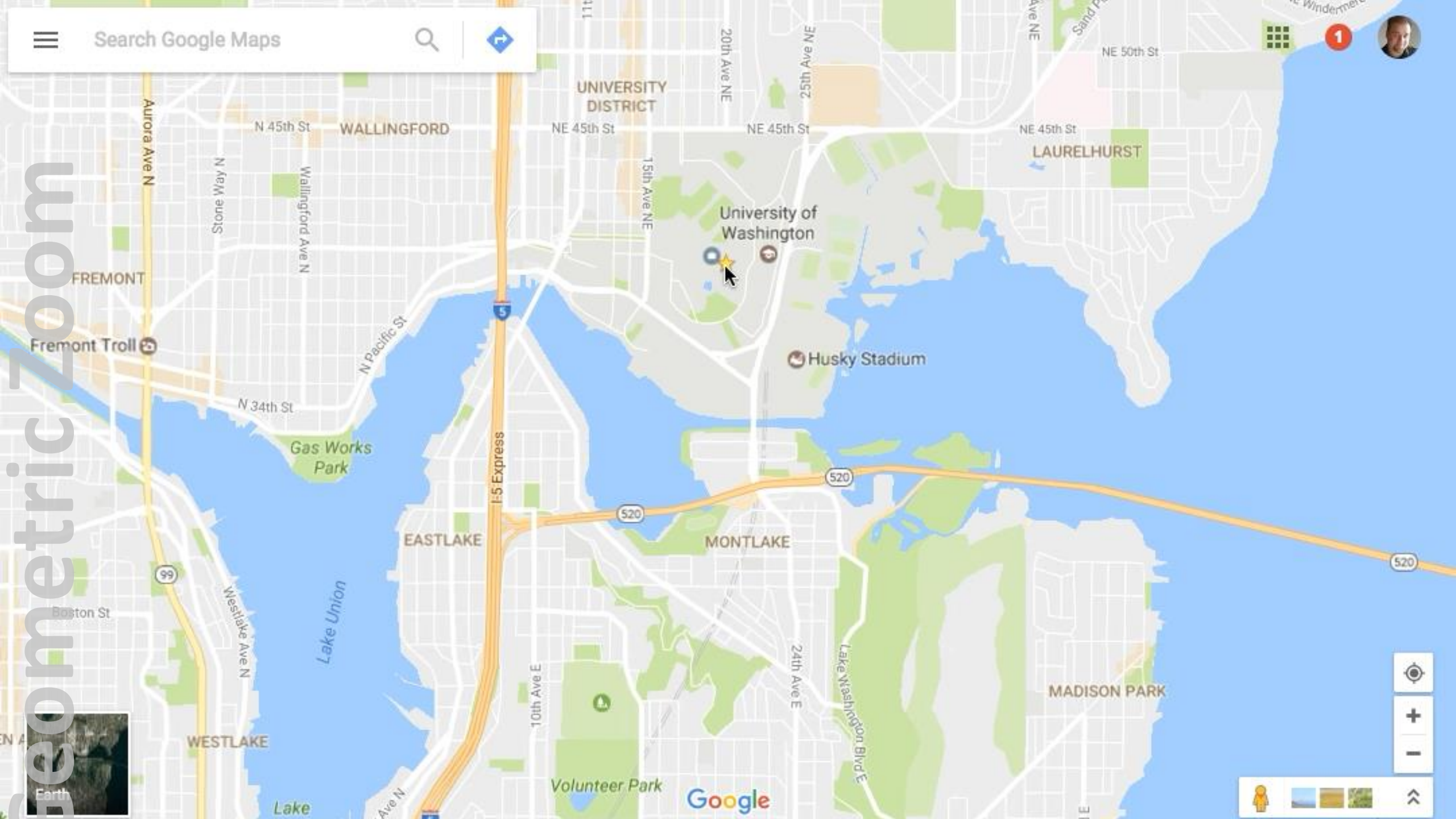


Search Google Maps



1

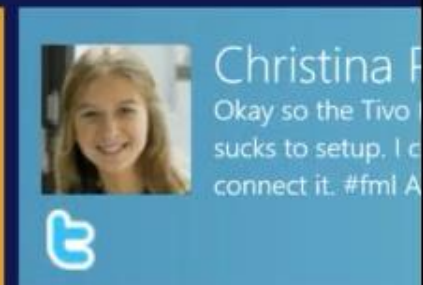
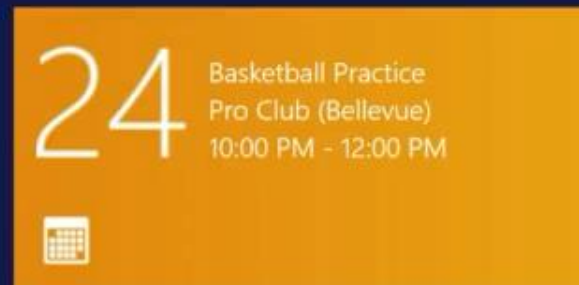
Geometric Zoom



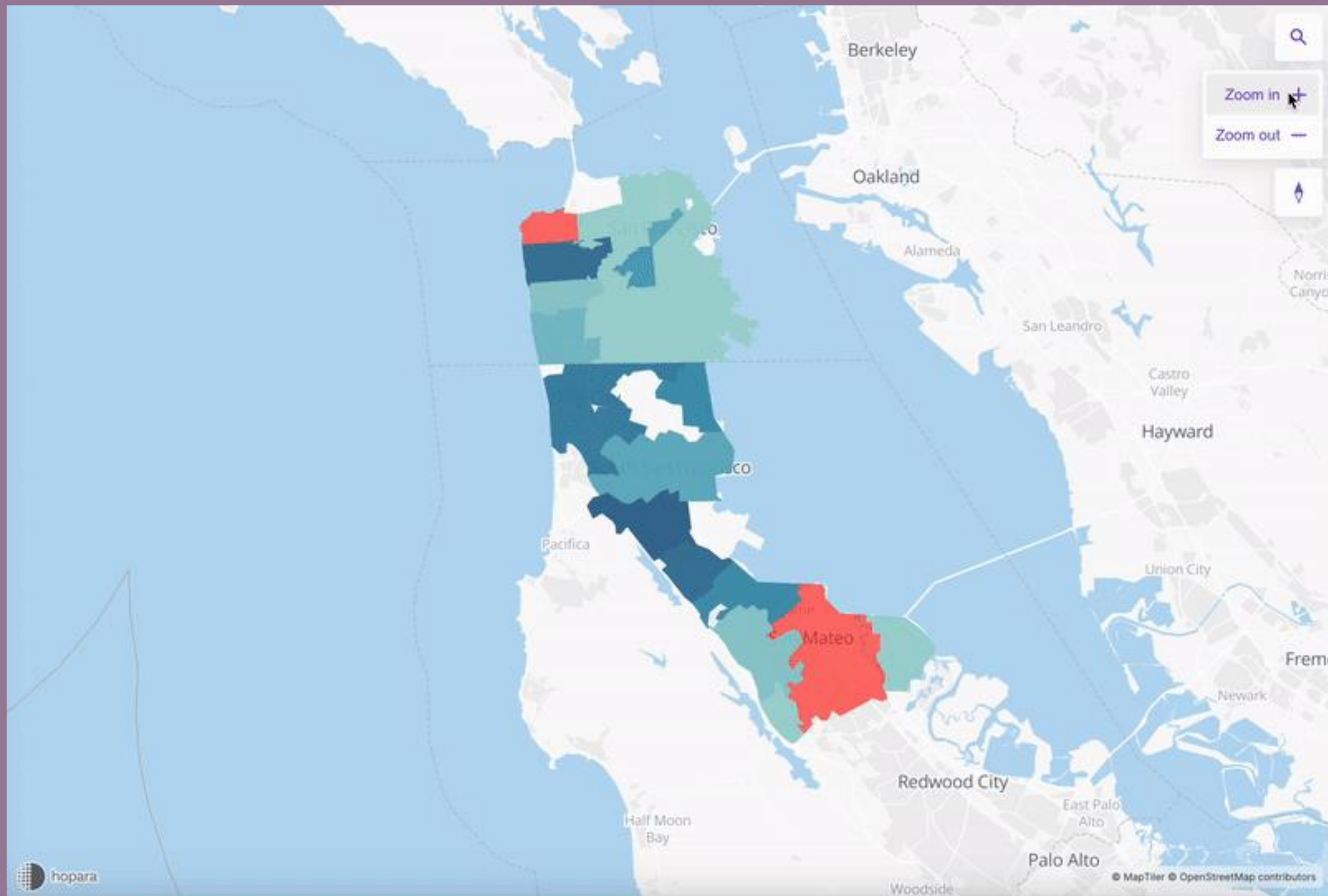
Google

Start

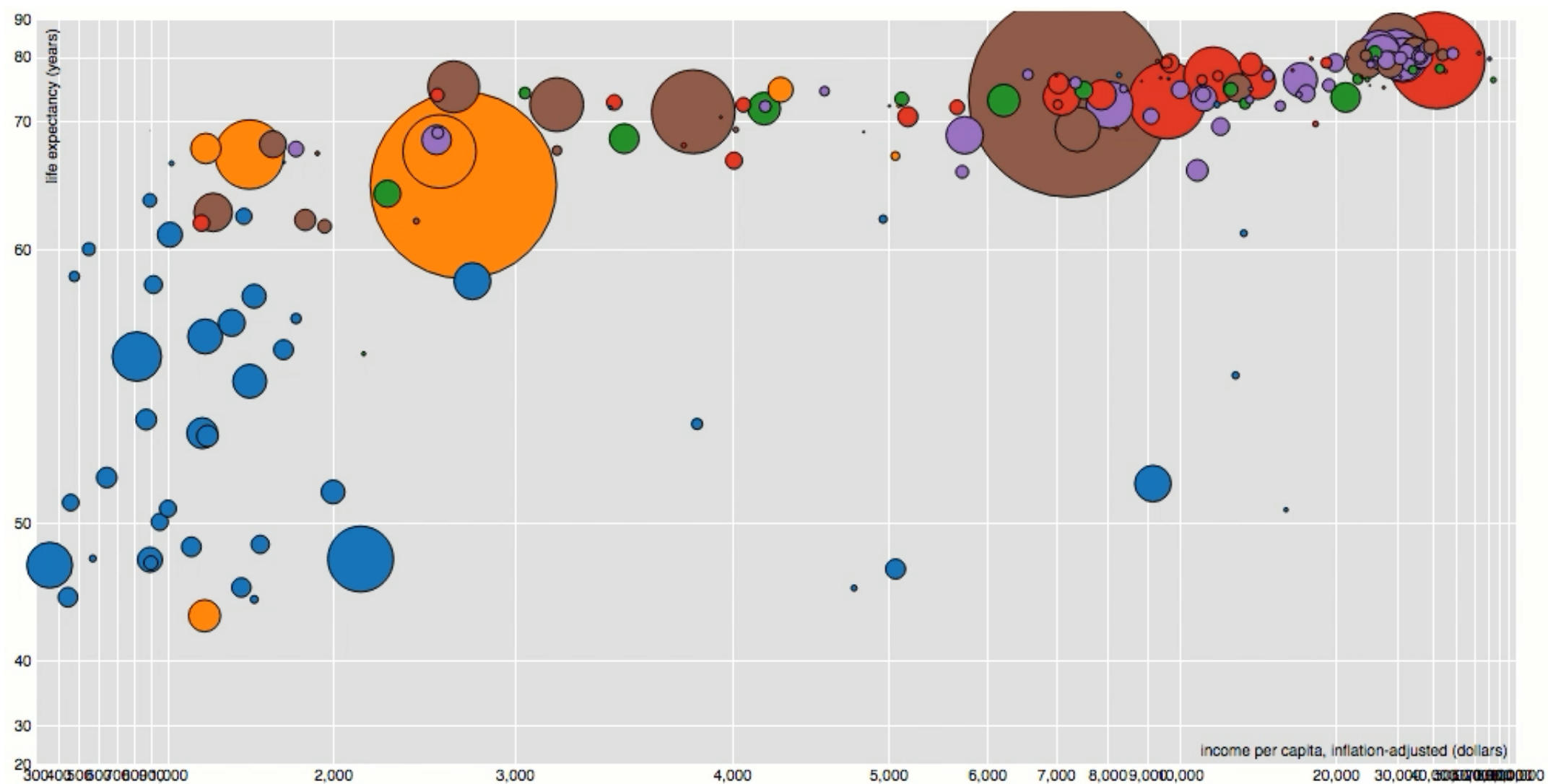
Alex
Smith 



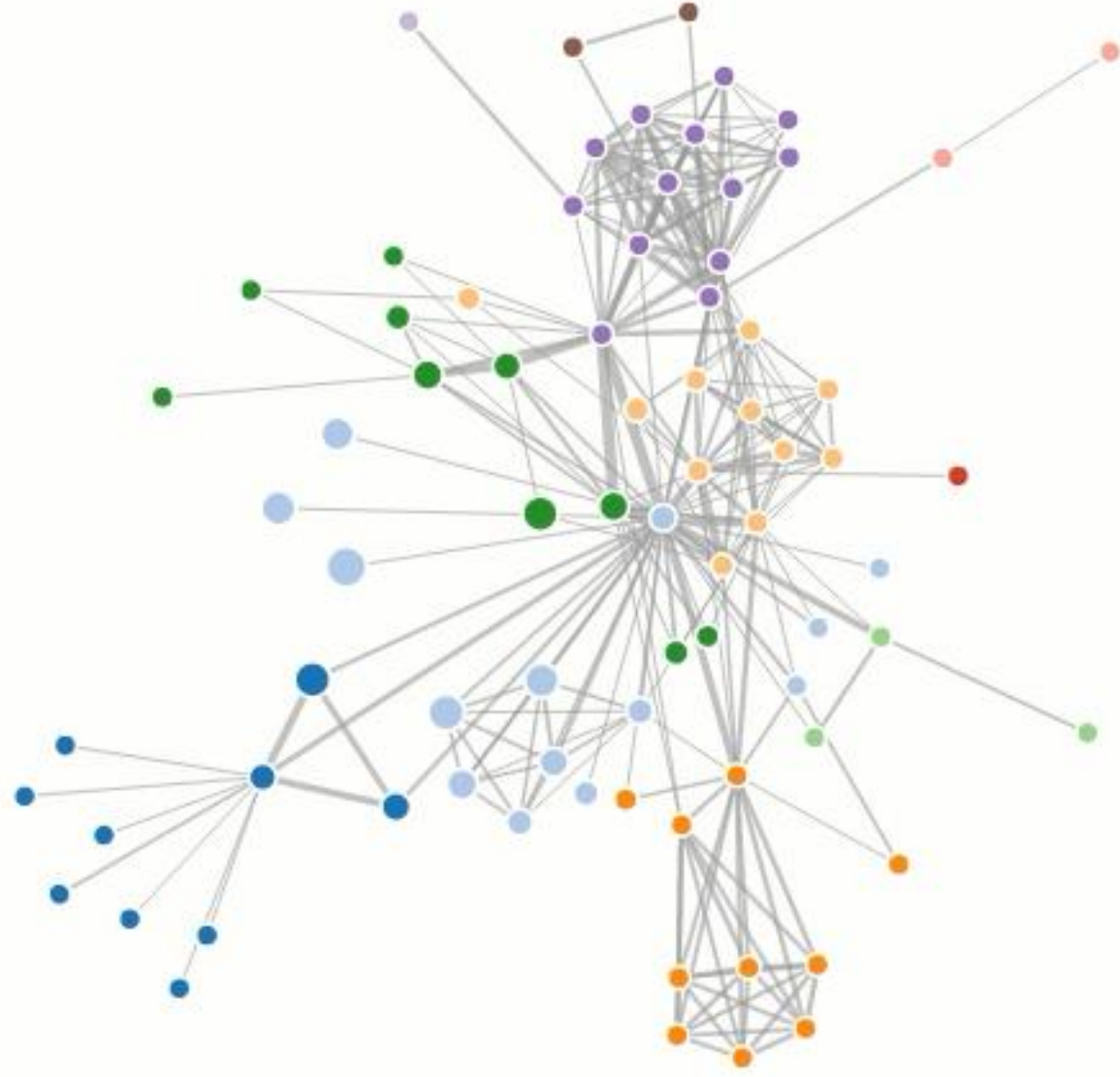
Semantic Zoom



Cartesian Distortion



Fisheye Distortion



Data Specification

Visualize

Filter

Sort

Derive

Aggregate

Separate

...

View Manipulation

Select

Navigate

Coordinate

Organize

Facet (juxtapose,
partition, superimpose)

Embed

...

Advanced

Brushing and Linking
(Highlighting)

Dynamic Queries (Filtering)

Overview + Detail

Focus + Context

Panning and Zooming

Distortion-based Views

...

You can invent new ones!

Most visualizations are interactive.

**Good visualizations are task
dependent and match user needs.**

**Fundamental interaction techniques
enable visual exploration.**

Quick Break
until x:xx pm

Activity

Within your project groups:

- 1 List out what your tasks might be
- 2 Brainstorm about what **interactions** might best support users in their tasks
- 3 Quickly **sketch** out how you might use these interactions in rough concepts – no need to show real data
- 4 Add sketches on a Google Slide
- 5 **Present** your ideas to the class

20 minutes

Next Week

Collaborative Critique!

Teams will pair up to **consult** with each other on how to improve progress on their final projects.

Each team needs to **come prepared with a brief overview (8-10 min)** of their project and current challenges.

Consulting teams will **provide recommendations** and share them with the partner team.